International Journal of Refrigeration 191 (2026) 107098

Contents lists available at ScienceDirect

International Journal of Refrigeration

journal homepage: www.elsevier.com/locate/ijrefrig

INSTITUT INTERNATIONAL DU FROID
INTERNATIONAL INSTITUTE OF REFRIGERATION

Review

Single-phase immersion coolants in data centers: From classification and performance to scenario-driven selection and perspectives☆,☆☆

Yaran Liang, Shuangquan Shao *

School of Energy and Power Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China

ARTICLE INFO

Keywords:
Single-phase immersion cooling
Data center
Coolant
Thermal management
Heat dissipation performance
Transient thermal buffering performance

ABSTRACT

With the rapid development of high-power-density data centers and global carbon neutrality drive, single-phase immersion cooling (SPIC) has become a key high-efficiency thermal management technology owing to its outstanding heat transfer performance and low energy consumption. As the core medium of SPIC systems, coolants directly play a critical role in promoting energy saving, consumption reduction, and sustainable development of data centers. This paper provides a comprehensive and systematic review of SPIC coolants. A refined classification framework is established, covering both traditional dielectric coolants and emerging functional coolants. The key thermophysical properties, electrical characteristics, material compatibility, safety, and environmental performance of various coolants are systematically summarized. Distinctively, by incorporating transient thermal buffering performance under dynamic thermal loads and combining it with steady-state heat dissipation capacity, it is possible to evaluate the thermal performance of coolants more comprehensively. A multi-dimensional comparison framework is constructed to reveal the trade-offs and complementarities among different coolants, providing clear guidance for scenario-specific selection. Furthermore, current technical challenges and future development directions toward high-efficiency, low-cost, and eco-friendly SPIC coolants are discussed. This review not only supports the selection, optimization, and innovation of SPIC coolants in data centers, but also provides a useful reference for the research and engineering application of high-performance dielectric fluids.

1. Introduction

1.1. Background and objectives

With the rapid adoption of artificial intelligence, and cloud computing, global data center computing demand has increased exponentially (China Academy of Information and Communications Technology, 2025), as shown in Fig. 1. Energy consumption in data centers has therefore become a major concern. As the primary energy-consuming subsystem, cooling systems account for up to 40% of total energy consumption (Liang et al., 2023). In traditional data centers, cooling already represents a significant share of energy consumption, and this issue becomes even more severe in high power density scenarios, placing strong pressure on energy efficiency improvement and carbon emission reduction.

To address this challenge, data center energy efficiency standards continue to be upgraded worldwide. Driven by “Dual Carbon” goals, China’s “Special Action Plan for Green and Low-Carbon Development of Data Centers” (National Development and Reform Commission of China, 2024) requires that, by the end of 2025, the Power Usage Effectiveness (PUE) of newly constructed, renovated, or expanded large and hyperscale data centers be reduced to 1.25 or below, and that the PUE of national hub data center projects not exceed 1.2. The EU’s “Code of Conduct on Data Centre Energy Efficiency” (Acton et al., 2025) and relevant US energy efficiency standards (Lawson et al., 2026) have also been tightened. Against this backdrop, liquid cooling technology has emerged as a key pathway for the green and low carbon transformation of data centers with the highly efficient heat dissipation capabilities.

Current data center liquid cooling technologies mainly include cold plate, immersion, and spray cooling (Cai and Gou, 2026). A comparison of liquid and air-cooling technologies is presented in Table 1. Traditional air-cooling is restricted by the low thermal conductivity (λ) and specific heat capacity (c_p) of air, limiting the rack power density to 15–20 kW (Al

☆ Submitted for review and possible publication in *International Journal of Refrigeration*
☆☆ Version date: July 29, 2026

* Corresponding author at: 1037 Luoyu Road, Wuhan, China.
E-mail address: shaoshq@hust.edu.cn (S. Shao).

https://doi.org/10.1016/j.ijrefrig.2026.107098
Received 23 June 2026; Received in revised form 29 July 2026; Accepted 18 August 2026
Available online 19 August 2026

Nomenclature

Symbols

c_p	Specific heat capacity, kJ/(kg·K)
P_p	Pumping power, W
q_v	Volumetric-specific sensible heat, MJ/m^3
r	Latent heat of vaporization, kJ/kg
Re	Reynolds number
Ra	Rayleigh number
T	Temperature, °C
T_a	Average CPU temperature, °C
ΔP	Pressure drop, Pa
ΔT	Temperature difference, °C

Abb-reviations

Alu	Aluminum alloys
CPU	Central processing unit
EFL	Electronic fluorinated liquid
FOM	Figure of merit
GPU	Graphics processing unit
GWP	Global warming potential
HBN	Hexagonal boron nitride
HDTMS	Hexadecyl trimethoxy silane
HFC	Hydrofluorocarbon
HFE	Hydrofluoroether
HFO	Hydrofluoroolefin
MO	Mineral oil
n.d.	Not determined
NE	Natural ester
ODP	Ozone depletion potential
PAO	Poly alpha olefins
PCB	Printed circuit board
PFAM	Perfluoroamine
PFAS	Per- and polyfluoroalkyl substance
PFAA	Perfluoroalkyl amine
PFAE	Perfluoroalkyl ether
PFC	Perfluorocarbon
PFO	Perfluoroolefin
PFPE	Perfluoropolyether
PUE	Power usage effectiveness
PVC	Polyvinyl chloride
SIO	Silicone oil
SPIC	Single-phase immersion cooling
SYO	Synthetic oil
TDP	Thermal design power
TPIC	Two-phase immersion cooling
UCO	Used cooking oil
UHFE	Unsaturated hydrofluoroether
VCO	Virgin coconut oil
WIC	Water immersion cooling

Greek symbols

ρ	Density, kg/m^3
β	Coefficient of thermal expansion, 1/K
ν	Kinematic viscosity, mm^2/s
μ	Dynamic viscosity, mPa·s
λ	Thermal conductivity, W/(m·K)
δ_T	Temperature uniformity index, °C

Kez et al., 2025) and easily causing local hotspots for high-power chips (Zou et al., 2025). Cold plate cooling achieves high reliability via indirect interface heat transfer but suffers from high thermal resistance, long heat transfer paths, and leakage risks (Huang et al., 2023). Spray cooling has simple structure and high heat dissipation efficiency (Yuan et al., 2024). However, it faces poor temperature uniformity, potential splashing issues, and strict atomization control requirements.

In contrast, immersion cooling submerges electronics directly in coolant, eliminating interfacial thermal resistance and providing excellent temperature uniformity and space utilization (Liu et al., 2025). Although two-phase immersion cooling (TPIC) offers higher thermal efficiency, it is characterized by extreme system complexity, and high maintenance costs (Pambudi et al., 2022). Single-phase immersion cooling (SPIC) offers high stability and excellent cooling capacity. It can reduce PUE to below 1.1. By balancing efficiency with scalability, SPIC has become a preferred solution for ultra-high power density data centers.

Existing literature has provided a relatively comprehensive review of SPIC technology in data centers (Pambudi et al., 2022; Ji et al., 2026; Zhou et al., 2024; Zheng et al., 2025; Yang et al., 2026; Lua and Wang, 2026; Zhang et al., 2025). The focus has been on technical principles, system design, heat transfer characteristics, and engineering applications. However, most reviews prioritize system-level analysis and cooling technology comparisons while neglecting in-depth investigations of coolants. Representative coolant-related review studies are summarized in Table 2. As shown in the comparison, current SPIC coolant research remains unsystematic. Existing literature lacks detailed elaboration on the development evolution, systematic classification, and key challenges of immersion coolants. It merely provides simple macroscopic categorizations of conventional oil and fluorocarbon coolants without clarifying their inherent structural differences, and fails to cover emerging coolants including nanofluids and conductive fluids. Some studies exhibit unbalanced research depth across different coolants, with sufficient analysis of oil coolants but superficial discussion of fluorocarbon coolants. Furthermore, current evaluation systems mainly rely on fragmented steady-state thermophysical, electrical, and environmental indicators, ignoring the transient thermal response buffering capability of coolants under time-varying heat loads, which results in one-sided assessment of coolant comprehensive performance.

Targeting the above research gaps, this paper presents a comprehensive review centered on SPIC coolants. Different from previous system-oriented studies, this work focuses on coolants. Firstly, a refined classification system covering both traditional dielectric coolants and emerging functional coolants is established, based on a systematic review of the development progress of immersion coolants. Secondly, the multi-dimensional performance of various coolants is comprehensively compared in terms of thermophysical, electrical, and environmental characteristics. Importantly, the transient thermal buffering performance under dynamic heat loads is supplemented to achieve a more comprehensive evaluation of coolant thermal performance. Finally, the key challenges and future development directions of SPIC coolants are prospected. This study aims to improve the current coolant research system, provide effective guidance for engineering coolant selection, and facilitate the large-scale and low-carbon application of SPIC technology in high power density data centers.

1.2. Review methodology and structure

To systematically review research progress on SPIC coolants for data centers, this study conducted a literature search and screening in the relevant field to ensure the comprehensiveness and accuracy of the review. The search covered multiple authoritative databases, including Web of Science, ScienceDirect and Scopus, comprehensively retrieving research findings on SPIC coolants for data centers from 2016 to the present. The literature screening was carried out in two steps, as shown in Fig. 2a. The search strategy focused on the core theme by combining search terms, and using Boolean operators to ensure precise matches. After an initial screening, a total of 405 papers were identified. Subsequently, a further screening of these papers excluded 303 ineligible papers, including those containing misleading keywords or topics that did not match the search criteria. 102 papers were retained as the analytical foundation for the review. Statistical trends indicate that the volume of publications in this field has shown an overall upward trend over the past decade (as Fig. 2b).

To overcome limitations in database retrieval, this study adopted a snowball sampling approach to trace both cited and citing references from the initial 102 articles. This was further supplemented with relevant corporate reports, industry publications, and government policy documents, resulting in a total of 148 references included in this review. Although every effort was made to ensure comprehensiveness, new research findings in this field continue to emerge. We sincerely apologize for any potential omissions of high-quality literature.

2. Properties and classification of immersion coolants

SPIC coolants include several technical routes, such as hydrocarbon and organosilicon (oil) coolants, fluorocarbon coolants, modified nanofluids, and insulating coating-assisted conductive fluids, with new types continuously emerging. These coolants exhibit distinct thermophysical, chemical, electrical, safety, and environmental characteristics that determine their cooling adaptability in data center scenarios.

2.1. Core properties of immersion coolants

As shown in Fig. 3, the selection of SPIC coolants for data centers must consider thermal, chemical, electrical, safety, and environmental

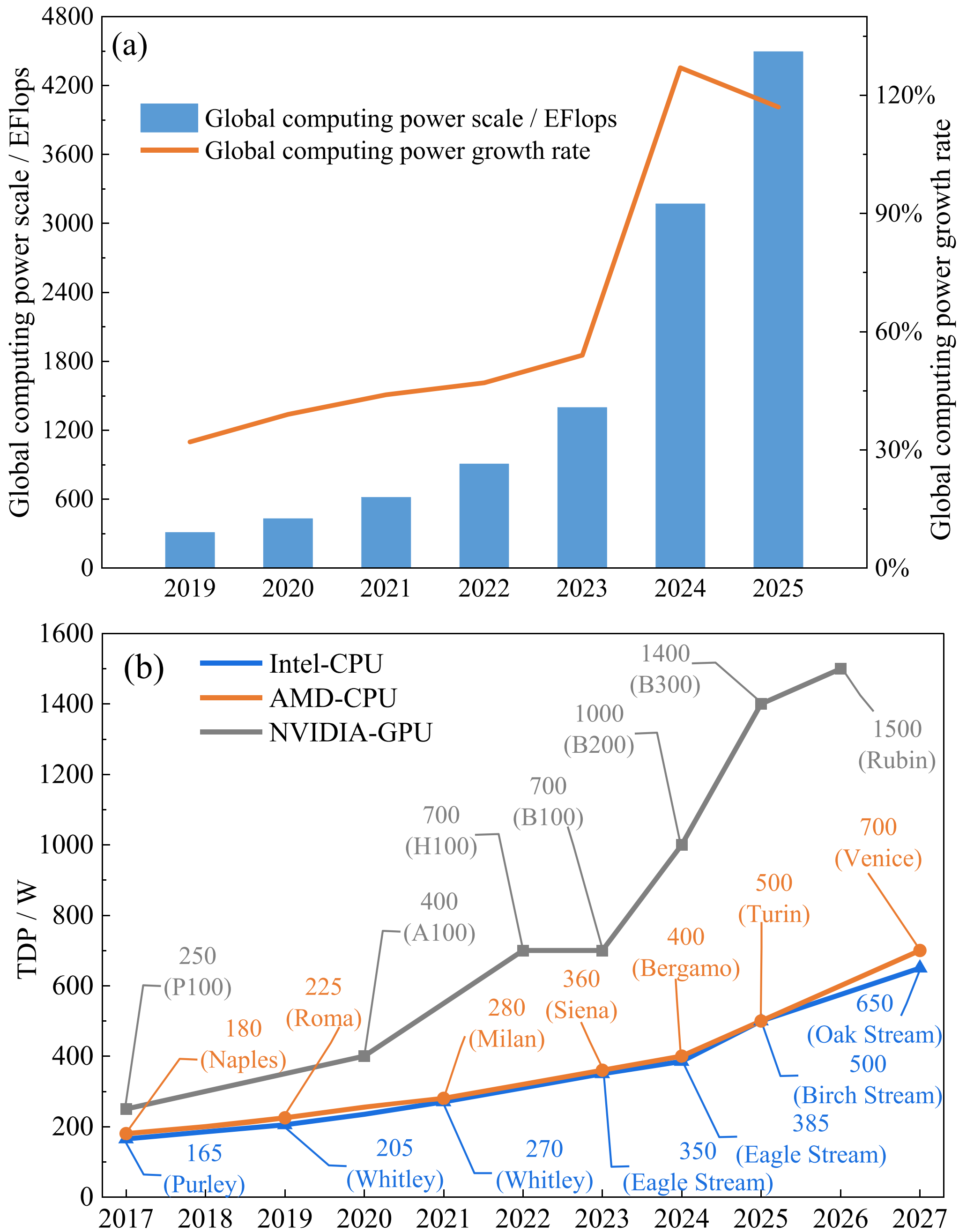

Fig. 1. Trends in computing power demand: (a) Trend chart of global computing power scale and growth rate (China Academy of Information and Communications Technology 2025), (b) Trend chart of TDP (Intel 2026; AMD 2026; NVIDIA 2026; Lv, 2026; Intel 2026; ARES-WHI0 2026).

properties (Azarifar et al., 2024). Thermal properties directly affect cooling efficiency and system stability (Hu and David, 2024), and also govern the transient thermal buffering capacity of coolants under variable heat loads. The dynamic temperature regulation performance originates from the sensible heat absorption of coolants, which can be characterized by mass- and volume-based indicators. Mass-based thermal buffering capacity is mainly determined by c_p, whereas volumetric metrics further incorporate density (ρ) and adopts volumetric-specific sensible heat (q_v) to better reflect the ability to counteract sudden rises in chip heat flux within confined cabinet spaces. Therefore, c_p and ρ are key factors governing the transient thermal buffering performance of coolants.

Since immersion coolants are in direct contact with electronic components, chemical properties such as compatibility, stability, and corrosiveness must be carefully evaluated (Zhang et al., 2025). Material compatibility, particularly whether the coolant reacts with or dissolves components, directly affects system feasibility and service life (Koster et al., 2023). In addition, because electronic systems require continuous operation, coolant insulation performance is critical. Volume resistivity and dielectric strength are key indicators of electrical safety. Signal compatibility must also be considered, where the dielectric constant plays an important role in transmission integrity. Finally, environmental impact and cost are essential considerations under carbon reduction targets and practical deployment requirements. An ideal immersion coolant must meet the technical specifications shown in Table 3. Of course, these indicators represent idealized targets that are difficult for current coolants to satisfy simultaneously. Nevertheless, they remain critical benchmarks for comparing different immersion coolants.

2.2. Classification of immersion coolants

Currently, immersion coolants for data center immersion cooling are mainly divided into hydrocarbon and organosilicon (oil) coolants and fluorocarbon coolants. In recent years, nanofluids and conductive fluids with insulating coatings have also attracted increasing attention. Fig. 4 provides a detailed classification of the four types of immersion coolant mentioned above.

Oil coolants are typically viscous at room temperature and include hydrocarbon and organosilicon fluids. They are characterized by high boiling points, low volatility, non-corrosiveness, low toxicity, environmental friendliness, and low cost. However, their flammability due to a finite flash point is a key safety concern. In addition, compatibility issues may arise from impurities, requiring strict long-term material compatibility evaluation.

Fluorocarbon coolants are organic compounds in which hydrogen atoms are partially or fully replaced by fluorine. They generally provide superior heat transfer performance, are non-flammable with no flash point, and exhibit long service life, chemical stability, and low viscosity. Owing to strong C-F bonds, they are highly inert and show excellent material compatibility, ensuring high system reliability (Wang et al., 2024; Wang et al., 2023; Zheng et al., 2024). However, the cost and GWP of fluorocarbon coolants are significantly higher than those of oil coolants.

As novel coolants in the SPIC systems, nanofluid coolants and conductive fluid coolants can significantly enhance heat transfer efficiency. A nanofluid is a colloidal suspension in which nanoparticles are uniformly dispersed in a base fluid such as water, oil, or organic fluids (Das et al., 2007). Adding highly thermally conductive nanoparticles to coolants not only increases the overall thermal conductivity but also enhances the convective heat transfer coefficient by increasing turbulence and improving heat exchange between the fluid and the heated surface. The large specific surface area of nanoparticles facilitates an effective connection between hot and cold zones, further improving heat conduction. Furthermore, to overcome the reliance of traditional coolants on insulating fluids, conductive fluid coolants combine insulating coating technology to effectively achieve the synergy of electrical insulation and efficient heat dissipation. However, their further promotion and application still face respective challenges.

3. Hydrocarbon and organosilicon (oil) coolants

Oil coolants are the most widely used and technically mature coolants in the SPIC field. Based on their source and synthesis method, they are primarily classified into four categories: mineral oils (MO), synthetic oils (SYO), silicone oils (SIO), and natural esters (NE). Table 4 summarizes the oil coolants currently used in data center SPIC systems and their properties.

Table 1
Comparison of data center cooling technologies (Wang et al., 2024; Kong et al., 2024).

Technology		Contact Mode	Coolant / Heat Transfer Medium	Heat Dissipation Performance	PUE	Retrofit & Maintenance Difficulty	Cost	Space Utilization	Technology Maturity	Application Scenarios
Air Cooling		Non-contact	Air	Poor; prone to local hot spots	1.5–1.8	Low	Low	Low	Very High	Low power density data centers
Cold Plate		Indirect	Deionized water, glycol solution, organic coolants	Good; with contact thermal resistance	1.1–1.25	Medium	Medium initial cost, low operating cost	Medium	High	High power density data centers
Immersion Cooling	Single-phase	Direct	Hydrocarbon, silicone, fluorocarbon coolants	Excellent; no contact thermal resistance; good temperature uniformity	1.05–1.1	High	High	High	Medium	Ultra-high power density data centers; AI clusters
	Two-phase	Direct	Silicone, fluorocarbon coolants	Optimal; high phase-change heat absorption efficiency	<1.05	Extremely High	Extremely High	High	Low	Supercomputers; Extreme-high power density pilots
Spray Cooling		Direct	Hydrocarbon, silicone, fluorocarbon coolants	Good; relatively poor temperature uniformity	<1.1	Medium	Low initial cost, high liquid consumption & operating cost	Medium	Medium	Low-to-medium power density data centers; Partial equipment heat dissipation

Table 2
Review of recent years' single-phase immersion coolants.

Ref	Section	Classification	Performance	Evaluation Criteria	Analysis Completeness
(Ji et al., 2026)	3.1 3.2	Oil coolants (mineral oils, synthetic oils, silicone oils) Fluorocarbon coolants (hydrofluorocarbons, perfluorocarbons, and hydrofluoroethers), Nanofluids	Thermophysical, Electrical, safety, Environmental	Thermal properties, toxicity, Global Warming Potential (GWP)	Targeted detailed introduction. Detailed exploration of oil-based coolant modification was conducted, while systematic discussion on fluorocarbon coolants and overall performance evaluation was lacking.
(Zheng et al., 2025)	2.1 2.2 2.3	Oil coolants (mineral oils, synthetic oils, silicone oils, vegetable oils) Electronic fluorinated liquids (hydrofluoroethers and perfluorocarbons)	Thermal, Electrical, Chemical, Environmental	Figure of Merit (FOM), Dielectric Constant, Dielectric Strength, Material Compatibility, Ozone Depletion Potential (ODP), GWP	Partial detailed introduction. Detailed analysis of oil-based coolants was performed, while the classification of fluorinated coolants was incomplete, and relevant comparative analysis was insufficient.
(Yang et al., 2026)	3.1	Oil coolants (mineral oils, synthetic oils, silicone oils) Fluorocarbon coolants (hydrofluorocarbons, perfluorocarbons)	Thermal, Electrical, Chemical, Environmental	FOM, Dielectric Constant, Dielectric Strength, Material Compatibility, ODP, GWP	Brief and superficial introduction. Only basic attribute description was conducted, lacking refined classification comparison, in-depth analysis and systematic performance evaluation.
(Lua and Wang, 2026)	3.1	Hydrocarbons (mineral oils, synthetic esters), fluorinated liquids, Nanofluids	Thermophysical, chemical	Thermal properties, Material Compatibility	Brief introduction. Only the most common coolant types were briefly summarized with incomplete classification and superficial performance analysis, while a preliminary exploration of nanofluid coolants was conducted.
(Zhang et al., 2025)	3.1 3.2 3.3	Oil coolants (mineral oils, synthetic oils, silicone oils) Fluorocarbon coolants (chlorofluorocarbon, hydrochlorofluorocarbon, hydrofluorocarbon, hydrofluoroether)	Thermal, Electrical, Chemical, Secure	FOM, Material Compatibility, volume resistivity, dielectric strength, Dielectric Constant, ODP, GWP	General overview introduction. Multiple coolants were covered, but incomplete classification criteria were adopted, without in-depth exploration and performance comparison.

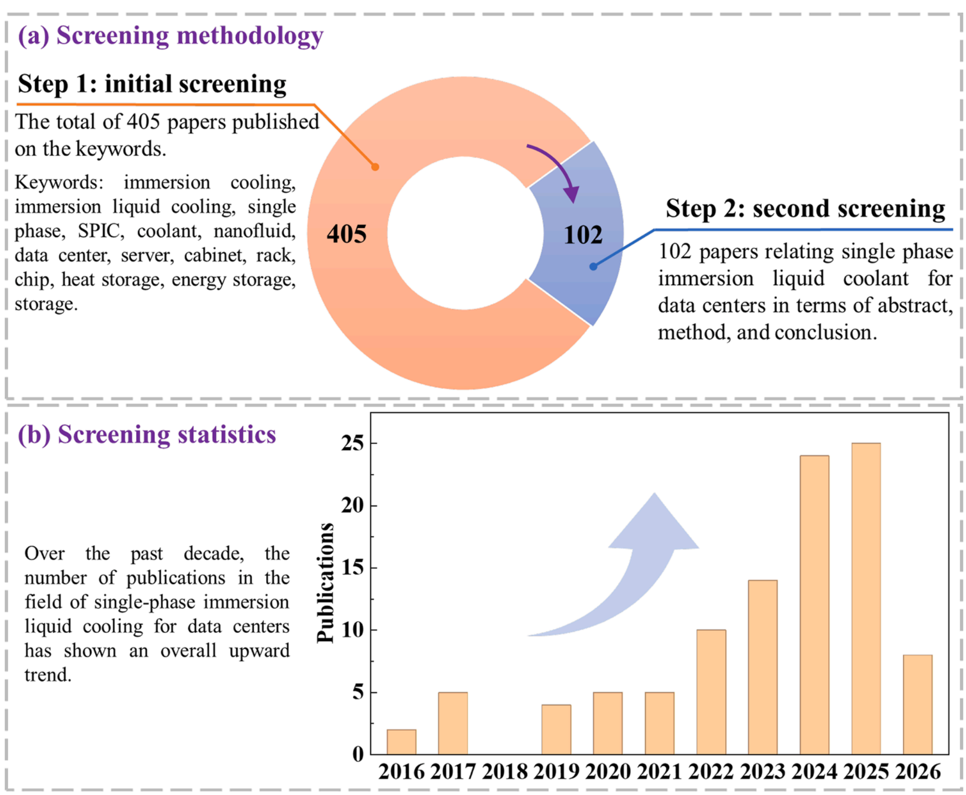

Fig. 2. Schematics of screening methodology and statistics results.

3.1. Mineral oil

MOs are colorless, odorless mixtures of liquid hydrocarbons refined from petroleum. Their relatively low cost among oil coolants makes them the most suitable option for large-scale applications. Compared to traditional air cooling, using MO for SPIC has shown promise as a viable alternative cooling solution for data centers (Eiland et al., 2014), achieving a PUE of 1.03 to 1.17 (Eiland et al., 2017).

In terms of thermal properties, Pambudi et al. (Agung Pambudi et al., 2022) found that MO exhibits superior cooling performance and cooling

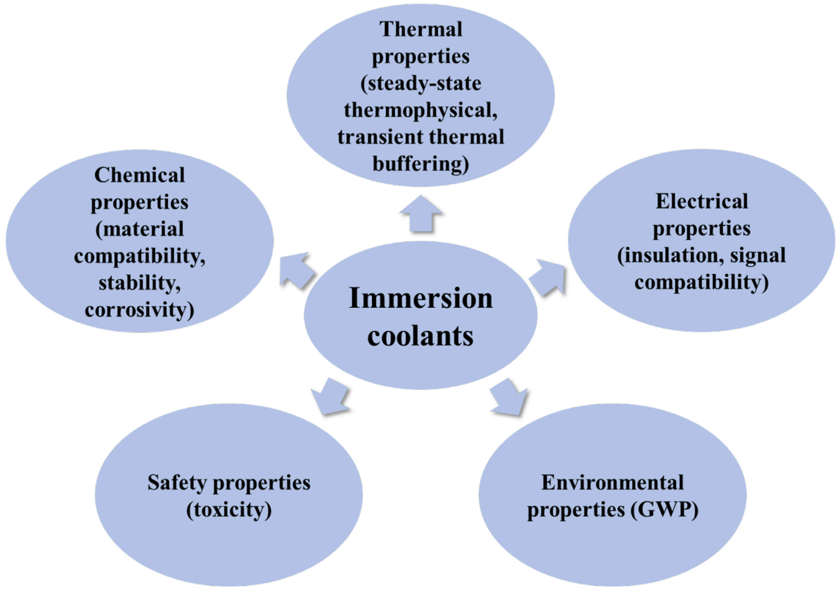

Fig. 3. Core properties of immersion coolants.

Table 3
Technical specifications for ideal immersion coolant.

Number	Property	Requirements
1	Flow characteristics	Low surface tension and low viscosity: Kinematic viscosity (ν) < 5×10^{-5} m^2/s (Ministry of Industry and Information Technology of the People's Republic of China, 2021)
2	Phase-change characteristics	High boiling point: boiling point > 100 °C
3	Thermal performance	Excellent heat transfer properties: $c_p \geq 0.96$ kJ/(kg·K) and $\lambda \geq 0.06$ W/(m·K)
4	Material compatibility	Good material compatibility and no corrosion of electronic components (Ministry of Industry and Information Technology of the People's Republic of China, 2021)
5	Chemical stability	High chemical stability, a high flash point or no flash point, and non-flammability
6	Electrical insulation	Excellent insulation properties: volume resistivity > 1×10^{12} Ω·cm and dielectric strength > 24 kV (2.54 mm gap) (Ministry of Industry and Information Technology of the People's Republic of China, 2021)
7	Signal compatibility	Good signal compatibility: the dielectric constant is < 2.5 (at 1 kHz), ensuring that high-frequency electronic components and connectors immersed in the coolant do not suffer significant loss of signal integrity (Zhang et al., 2021)
8	Toxicity	Low toxicity: median lethal dose (LD50) > 2000 mg/kg (Ministry of Industry and Information Technology of the People's Republic of China, 2021)
9	Environmental performance	Environmentally friendly: ODP = 0 and GWP < 250 (Bulinski et al., 2020)

effectiveness compared to virgin coconut oil (VCO). In fact, among oil coolants, viscosity has a more significant impact than other thermophysical properties. A lower dynamic viscosity (μ) means that the coolant consumes less power. Wang et al. (Wang et al., 2024) found that MO has a relatively low μ, effectively enhancing the heat transfer capacity. However, MO has a higher viscosity than SYO, with lower c_p and λ, resulting in limited heat transfer performance. Therefore, it is typically suitable for data centers with medium to low power densities. Although most MOs have a higher ρ, their low c_p results in a slightly lower q_v. While c_p of most MOs is below 2.0 kJ/(kg·K), SCL6 has a high c_p of 2.35 kJ/(kg·K), giving it the optimal q_v. Accordingly, it stands out as the MO with the greatest transient thermal buffering potential.

MO contains various impurities, poses a higher risk of oxidation and acid corrosion, and has poor long-term reliability (Shah et al., 2015). Therefore, testing their material compatibility is crucial. Pambudi et al. (Agung Pambudi et al., 2022) and Chauhan et al. (Chauhan et al., 2021) conducted durability tests on MO. No significant damage was observed in the components after prolonged immersion in MO. However, Shah et al. (Shah et al., 2019; Shah et al., 2021; Shah et al., 2016) found more micropores on PVC surfaces and molecular aggregation, leading to coating hardening. This is reflected in the material's elastic modulus. Excessively high elastic modulus renders the material stiff and brittle, potentially causing insulation failure, and solder crack initiation. These defects further lead to abrupt impedance changes and signal transmission attenuation. In contrast, an excessively low elastic modulus makes the material prone to creep deformation, degrading long-term insulation reliability. This will lead to poor contact and structural instability, causing fluctuations in signal impedance and degraded transmission integrity. Therefore, the hardening of the PVC coating plays a positive role in suppressing micro-bend loss, thereby helping to improve its signal compatibility. However, this benefit comes at the expense of material toughness and long-term reliability. In contrast, PCBs and passive components showed no significant changes in thermomechanical or electrical properties, indicating good compatibility. However, MO may corrode adhesives, inks, and markings, complicating equipment identification during maintenance, and accumulated debris or particles can cause pipeline blockages.

3.2. *Synthetic oil*

SYOs are hydrocarbons or esters produced through artificial synthesis. They include poly alpha olefins (PAO), gas to liquid base oils and synthetic esters. Compared with natural MOs, SYOs are synthesized with precisely controlled molecular structures, resulting from a more refined production process. Consequently, they contain fewer impurities and exhibit improved thermal stability, oxidation resistance, and material compatibility. Nevertheless, SYOs still face issues related to flash point.

SYOs exhibit excellent cooling performance due to their low viscosity (Hnayno et al., 2023; Shenyang Aojina Chemical Co. Ltd 2026). Compared to MO and SIO, the use of SYO reduces the average CPU temperature (T_a) by 3.9 °C and 2.7 °C, respectively. Meanwhile, the

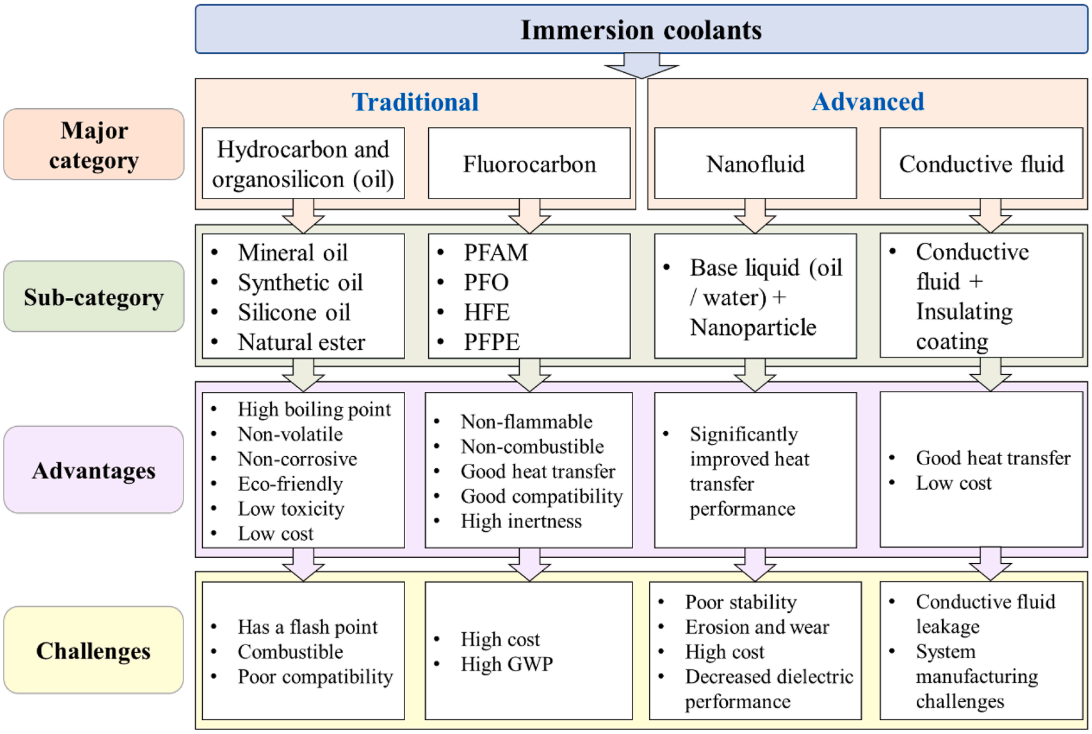

Fig. 4. Classification of immersion coolants.

pumping power (P_p) is reduced by approximately 1.58 times compared with MO and by >7.18 times compared with SIO (Khoshvaght-Aliabadi et al., 2025). Among FC-40, EC-130 and MO (Vagiakis et al., 2025), EC-130 absorbs 7.9% more heat than MO and 0.2% more than FC-40. Furthermore, Wang et al. (Wang et al., 2024) found that with SYO (Therminol VP-1), the server components exhibit the lowest temperature and minimal thermal resistance. Owing to its lowest viscosity, SYO results in the smallest pressure drop (ΔP).

SYOs not only have extremely low viscosity, but also possess higher λ and c_p, making them a mainstream choice for current medium-to-high power density SPIC systems. Compared with white MO, EC-100 demonstrates superior performance in maintaining lower server temperatures due to its higher λ and c_p (Shah et al., 2019). Compared with FC-40 and MO, EC-110 exhibits the most stable thermal behavior owing to its high c_p (Khoshvaght-Aliabadi et al., 2026). PAO-4 outperforms Noah 3000A. With a GWP of 0, PAO-4 is a more environmentally friendly immersion coolant (Zhang et al., 2024). The higher c_p of SYOs also results in a higher q_v. This enables SYOs to absorb more transient heat under identical temperature fluctuations, delivering strong thermal buffering performance against abrupt chip heat load surges in SPIC systems.

Research has shown that synthetic esters in SYOs have superior thermal properties, as well as better safety and environmental performance. BioLife 4, employed by Ruichek et al. (Ruichek et al., 2025), is a synthetic ester coolant produced from used cooking oil (UCO). It features high purity, excellent cooling performance, and good thermal, oxidation stability, and biodegradability (Finance News Network 2025). Its superior thermal properties enable the lowest T_a. Similarly, L759 is a synthetic ester dielectric coolant with low viscosity, high thermal conductivity, and high c_p, enabling efficient heat transfer. It also features a high flash point, high electrical resistivity, strong dielectric strength, and good environmental performance (Beau et al., 2024).

However, despite the exceptional thermophysical properties of SYOs, their reliability and material compatibility should be evaluated before application in SPIC data centers (Refai-Ahmed et al., 2022; Bhandari et al., 2022). Ramdas et al. (Ramdas et al., 2019) found that prolonged immersion in EC-100 reduced the elastic modulus of PCB by 33.33%, improving fatigue resistance and extending service life. Bansode et al. (Bansode et al., 2024) and Suthar et al. (Suthar et al., 2024) studied EC-100 and PAO-6 on TG-400 substrates and found both reduced the elastic modulus, with effects becoming more pronounced as temperature increased. For L759, Vaerenbergh et al. (Van Vaerenbergh et al., 2025) found that its acid value increased exponentially, while its c_p was significantly affected. Its dielectric constant remained stable, but electrical resistivity and breakdown strength decreased notably. However, Beau et al. (Beau et al., 2024) reported adhesive dissolution and incompatibility with label materials, indicating the need for compatible sealing materials. Therefore, selective compatibility with sealing materials remains a key issue requiring optimization.

3.3. Silicone oil

Similar to SYOs, SIOs are also obtained through artificial synthesis. Unlike MOs and SYOs, SIOs are not hydrocarbon coolants. Compared with them, SIOs have a higher flash point. However, the reduced flammability of SIOs is accompanied by flow difficulties, as flash point and viscosity tend to vary in the same direction. Therefore, a proper balance between these two properties must be struck. Furthermore, SIOs are susceptible to hydrolysis and oxidative deposits, which can affect contact performance.

High-viscosity SIO significantly increases the flow resistance and reduces the convective heat transfer capacity of SPIC systems (Sun and Liu, 2026; Matsuoka et al., 2017). Consequently, ΔP using SIO is 4.5 and 8.4 times higher than those using MO and SYO (Wang et al., 2024). The modified SIO developed by China Runhe Materials (Runhe Materials Technology Co. Ltd, 2025) demonstrates significant potential due to lower viscosity, lower cost, and better heat transfer performance. However, SIO generally has lower c_p than hydrocarbon coolants, leading to weaker transient thermal buffering capability in SPIC applications.

There has been little research on the material compatibility and reliability of SIO, and no relevant literature has been found in the data center field. To address this gap, the authors expanded the scope to

Table 4
Hydrocarbon and organosilicon (oil) coolants and properties.

Category	Name	Company	Boiling point / °C	Flash point / °C	Kinematic viscosity ν/mm^2·s^{-1}	Dynamic viscosity μ/mPa·s	Specific heat capacity c_p/ kJ·kg^{-1}·K^{-1}	Thermal conductivity λ/W·m^{-1}·K^{-1}	Density ρ/kg·m^{-3}	Volumetric-specific sensible heat q_v/MJ·m^{-3} ($\Delta T = 20$ °C)	Dielectric strength /kV·mm^{-1}	Dielectric constant	GWP	Ref
MO	Natural MO				14.13	12	1.67	0.13	849	28.36				(Sun and Liu, 2026)
MO					22.5	19.6	1.863	0.134	871.3	32.46				(Kim et al., 2025)
MO		CNPC			9.8	8.12	1.893	0.11	832.5	31.52				(Wang et al., 2024)
MO	white MO				16.02	13.6	1.67	0.13	849.3	28.37				(Shah et al., 2019)
MO	white MO	Macklin		185	6.6	5.6	2.2	0.13	849	37.36				(Luo et al., 2022)
MO	SCL6			154	6.01	5.1	2.35	0.14	851	40.00	31.2	2.05		(Pan et al., 2024)
SYO	PAO-2			159	5.1	4.1	2.3	0.15	795.7	36.60	17.2			(Guo et al., 2021)
SYO	PAO-4			219	21.5	17.6	2.042	0.153	819	33.45		2.2	0	(Kim et al., 2025)
SYO	PAO-6			246	35.6	29	2.377	0.145	814	38.70			0	(Zhang et al., 2024)
SYO	AC-100				11.3	9	2.137	0.14	795	33.98	30	2.1		(Kelly et al., 2021)
SYO	AC-200				12.4	10	2.137	0.16	807	34.49	30	2.21		(Kelly et al., 2021)
SYO	BC-888				15.9	13	2.133	0.14	817	34.85	20	2.2		(Kelly et al., 2021)
SYO	EC-100	Engineered Fluids		185	26	22	2.133	0.14	846	36.09	20	2.2		(Kelly et al., 2021)
SYO	EC-110	Engineered Fluids		203	8.2	6.8	2.212	0.136	833	36.85	30	2.1	0	(Li et al., 2023; Lamotte-Dawaghreh et al., 2024)
SYO	EC-120				9.5	8	2.133	0.14	846	36.09	30	2.1		(Kelly et al., 2021)
SYO	EC-130				5.9	5	2.133	0.14	846	36.09	30	2.2		(Kelly et al., 2021)
SYO	BINGYI 797	Tier	300		7.7	6.05	2.051	0.360	786	32.24			0	(Li et al., 2025; Liu et al., 2024)
SYO	VP-3		243		2	1.87	1.698	0.115	923	31.35			0	(Li et al., 2025)
SYO	ILC-5				8.069	6.9	1.94	0.149	849	32.94				(Sun and Liu, 2026)
SYO	L759	Oleon NV		192	7.3	6.6	1.96	0.145	898	35.20	30.8	3.5		(Van Vaerenbergh et al., 2025)
SYO	S5 X	Shell	285	200	9.8	7.9	2.274	0.142	806	36.66	16.8	2.02		(Kim et al., 2025; Hnayno et al., 2023)
SYO	SmartCoolant	Submer	295	159	6.4	5.1	2.26	0.14	799	36.11	≥ 20			(Hnayno et al., 2023; Taddeo et al., 2023)
SYO	ThermaSafe R	DCX	300	> 180	5.5	4.6	2.20	0.129	833	36.65	> 24			(Hnayno et al., 2023)
SYO	Therminol VP-1	Eastman	257		3.6	3.846	1.556	0.136	1060	32.99		3.35		(Wang et al., 2024)
SYO	MIVOLT DFK			> 250	75	72.6	1.902	0.147	968	36.82	> 30			(Pan et al., 2024)
SYO	MIVOLT DF7				16.4	15	1.91	0.129	916	34.99				(Beau et al., 2024)
SYO	BioLife 4	TotalEnergies		145	4	3.1	2.235	0.138	771	34.46				(Ruichek et al., 2025)
SIO	XIAMETER 561	Dow		> 300	50.5	48.3	1.471	0.144	957.2	28.16		2.7		(Wang et al., 2024)
SIO					38.1	36.6	1.60	0.155	960	30.72				(Sun and Liu, 2026)
SIO	Dimethyl SIO		140		5	4.57	1.55	0.12	913	28.30		2.7		(Han et al., 2016)
SIO	PCM-10		> 200	> 170	10	9.3	1.6	0.12	930	29.76	> 27.5	2.6		(Runhe Materials Technology Co. Ltd 2025)
SIO	DC-550				84	78.46	1.5	0.13	934	28.02				(Zhang et al., 2026)
SIO	PSF-20 cSt				20	19	1.6	0.14	950	30.40	25	2.68		(Kelly et al., 2021)
SIO	PSF-50 cSt				50	48	1.5	0.15	960	28.80	25	2.71		(Kelly et al., 2021)
SIO	PSF-100 cSt				100	97	1.5	0.16	970	29.10	25	2.71		(Kelly et al., 2021)
NE					72.7	67	1.994	0.183	922.1	36.77	45 ∼ 55			(Kim et al., 2025)
NE	NatureCool2000				31	28.6	2.31	0.164	921	42.55				(Beau et al., 2024)
NE	RAPO	Zhuoyuan		322	33.35	30.7	2.165	0.168	920	39.84		3.1 ∼ 3.3		(Liu et al., 2024)
	EGEN 100R8	Motul		166	8.1	6.4	2.24	0.140	787	35.26				(Ruichek et al., 2025)
	SL 3568	Shell		265	43	34.9	2.00	0.139	811	32.44				(Ruichek et al., 2025)
	FES 822–6542	Fuchs		204	11.5	9.3	2.25	0.134	813	36.59				(Ruichek et al., 2025)

applications of immersion coolants in other fields. Han et al. (Han et al., 2016) found that after 180 days of immersion in dimethyl SIO, the optical transmittance and normalized photocurrent density of labeled concentrated photovoltaic solar cells remained essentially unchanged. Therefore, dimethyl SIO can be considered the optimal coolant for direct immersion cooling in concentrated photovoltaic systems.

3.4. Natural ester

In addition to the commonly used oil coolants mentioned above, NEs represent another category. NEs are ester compounds extracted from oilseeds, typically existing as triglycerides, and are mainly used in power transformers (Ding et al., 2025). NEs offer good biodegradability, low toxicity, environmental friendliness, and excellent thermal stability, and also have higher breakdown voltage than MOs, improving safety and reliability (Zhang et al., 2025). BioLife 4, derived from UCO-based NEs, exhibits superior biodegradability and environmental performance than other synthetic esters. The significant advantages of NEs have led to increased research interest in their use as immersion coolants.

NatureCool2000 (Beau et al., 2024), a NE applied in data centers, exhibits favorable thermophysical properties due to its relatively high c_p and λ. However, its viscosity is relatively high, requiring further evaluation. Furthermore, Kim et al. (Kim et al., 2025) found that NE resulted in the highest T_a and the largest ΔP due to its highest viscosity. In contrast to T_a, immersion in NE led to lower memory temperatures than in FC-40, PAO-4 and MO. This may be related to the high c_p and λ of NE. Similarly, the high λ and ρ of NEs combine to give them excellent q_v.

Given the limited application of NEs in data centers, the literature is expanded to their physical and dielectric properties. For instance, FR3-type NE, used in oil-immersed transformers, exhibits a higher flash point than Karamay No 25 MO, along with markedly superior electrical performance (Wang et al., 2025). Its dielectric properties degrade slowly under repeated breakdowns, with only 7.8% increase in dielectric loss and 8.2% decrease in resistivity, both lower than MO. RAPO-type NE (Liu et al., 2024) exhibits excellent heat dissipation and environmental performance. making NEs attractive for green data centers due to their biodegradability and sustainability. However, their high viscosity limits heat transfer performance. To address this, recent studies have focused on blending NEs with synthetic esters (Qian et al., 2025) to balance performance, environmental benefits, and cost.

4. Fluorocarbon coolants

Fluorocarbon coolants can be classified into four major categories based on whether the H atoms in the compound are completely substituted by F atoms and whether the compound is saturated: hydrofluorinated saturated compounds, hydrofluorinated unsaturated compounds, perfluorinated saturated compounds, and perfluorinated unsaturated compounds (Zhang et al., 2022). Each category can be further subdivided into specific types, as illustrated in Fig. 5. It adopts a chemical derivation perspective showing the structural relationships among them. Among these, the types that have been most extensively commercialized and reported in the literature for data center SPIC applications are perfluoroamines (PFAM), hydrofluoroethers (HFE), perfluoropolyethers (PFPE), and perfluoroolefins (PFO). Therefore, this review focuses on the aforementioned four types (PFAMs, HFEs, PFPEs, and PFOs). A summary of their key properties is presented in Table 5.

4.1. Perfluoroamine

PFAMs are obtained by introducing N atoms into perfluorocarbons (PFCs). PFAMs belong to the class of perfluorinated saturated compounds. Currently, the PFAMs used in data center SPIC systems are primarily the FC series developed by 3 M. Compared to oil coolants, PFAMs have a lower viscosity, resulting in superior heat transfer performance (Huang et al., 2023). Compared to other fluorocarbons, PFAMs have a higher ρ, resulting in superior transient thermal buffering performance. Their elevated q_v mitigates sharp heat flux spikes from chips under variable operating loads. Meanwhile, PFAMs have relatively low vaporization latent heat due to weak intermolecular forces in their perfluorinated structure. Their dielectric constants are very low (all < 2). They generally have high dielectric strength, indicating excellent insulation properties. However, it is worth noting that the GWP of PFAMs is excessively high (all > 5000), contributing significantly to the greenhouse effect and failing to meet environmentally friendly standards.

The excellent heat transfer and insulation properties of PFAMs have been effectively validated (Ni and Yu, 2024; Jin et al., 2025; Liu and

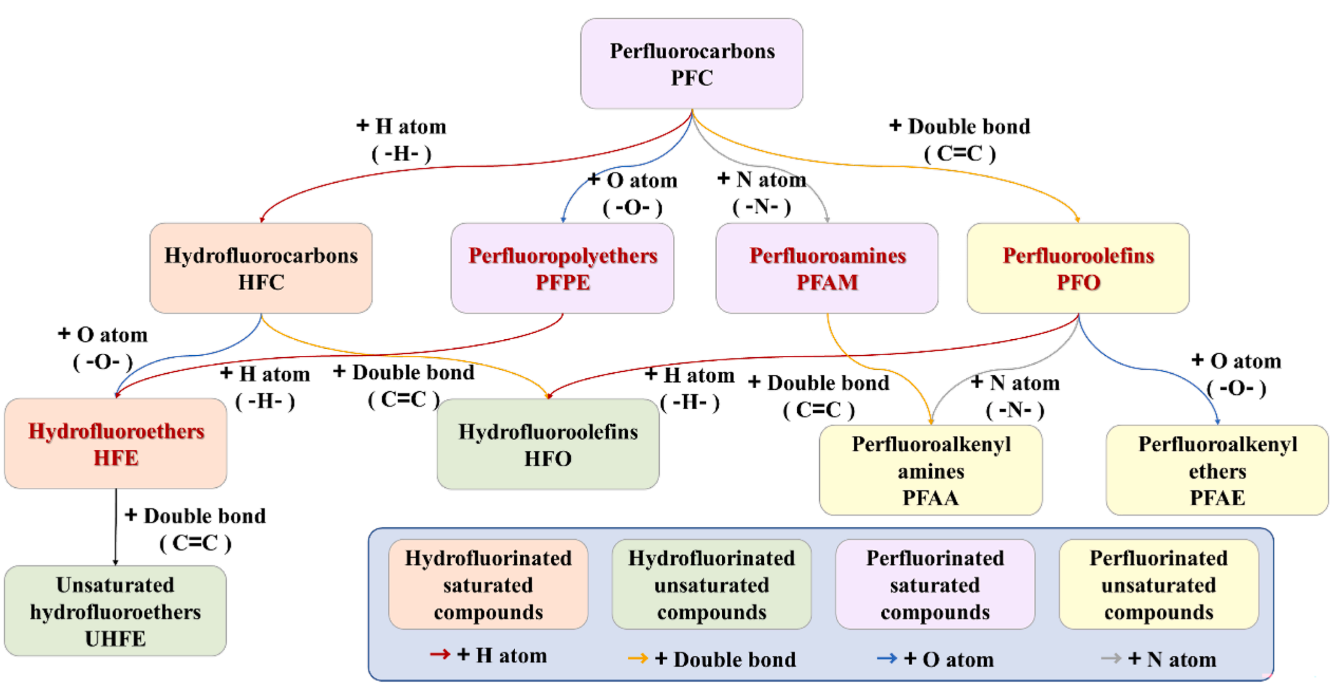

Fig. 5. Classification and correlation of fluorocarbon coolants.

Table 5
Fluorocarbon coolants and properties.

Category	Name	Company	Boiling point / °C	Kinematic viscosity ν/mm^2·s^{-1}	Dynamic viscosity μ/mPa·s	Specific heat capacity c_p/ kJ·kg^{-1}·K^{-1}	Thermal conductivity λ/W·m^{-1}·K^{-1}	Density ρ/kg·m^{-3}	Volumetric-specific sensible heat q_v/MJ·m^{-3} ($\Delta T = 20$ °C)	Latent heat of vaporization r/kJ·kg^{-1}	Dielectric strength /kV·mm^{-1}	Dielectric constant	GWP	Ref
PFAM	FC-40	3M	165	2.2	4.1	1.076	0.064	1860	39.23	69	18.1	1.9	> 5000	(Kim et al., 2025; Li et al., 2025)
PFAM	FC-43	3M	174	2.5	4.7	1.1	0.07	1860	40.92	70	16.54	1.9	> 5000	(Kelly et al., 2021)
PFAM	FC-70	3M	215	12	23.3	1.1	0.07	1940	42.68	69	15.75	1.98	> 5000	(3M 2019)
PFAM	FC-84	3M		0.6	1	1.1	0.06	1730	38.06		16	1.8		(Kelly et al., 2021)
PFAM	FC-770	3M		0.6	1	1.038	0.06	1793	37.22	85.9	16	1.9		(Kelly et al., 2021)
PFAM	FC-3283	3M	128	0.75	1.4	1.1	0.066	1820	40.04	78	16.93	1.9	> 5000	(Fan et al., 2025; Huang et al., 2025)
UHFE	SF-10	Chemours	110	0.6	0.87	1.252	0.076	1558	39.01	115	11.42	5.48	2.5	(Kim et al., 2025; Li et al., 2025)
HFE	Novec-7100	3M	61	0.38	0.5	1.183	0.069	1418.6	33.56	112	> 9.84	7.4	320	(Cheng et al., 2020; Rokonuzzaman et al., 2025)
HFE	Novec-7200	3M	76	0.41	0.6	1.22	0.075	1430	34.89	119	> 9.84	7.3	55	(Khoshvaght-Aliabadi et al., 2025; Zhang et al., 2022)
HFE	Novec-7300	3M	98	0.7	1.18	1.14	0.069	1660	37.85	102	> 9.84	6.1	200	(Khoshvaght-Aliabadi et al., 2025; Zhang et al., 2022)
HFE	Novec-7500	3M	128	0.6	0.98	1.151	0.062	1582	36.42	89	13.78	5.8	90	(Kim et al., 2025; Li et al., 2025)
HFE	Novec-7700	3M	167	2.52	3.1	1.04	0.064	1797	37.38	83	13.78	6.7	420	(Fu et al., 2025)
PFPE	HT110	Solvay	110	0.77	1.3	0.96	0.065	1710	32.83	69.09	15.75	1.92	> 5000	(Solvay 2024)
PFPE	HT135	Solvay	135	1	1.7	0.96	0.065	1720	33.02	69.09	15.75	1.92	> 5000	(Solvay 2024)
PFPE	HT170	Solvay	170	1.8	3.2	0.96	0.065	1770	33.98	69.09	15.75	1.94	> 5000	(Solvay 2024)
PFPE	HT200	Solvay	200	2.4	4.3	0.96	0.065	1790	34.37	62.79	15.75	1.94	> 5000	(Solvay 2024)
PFPE	HT230	Solvay	230	4.4	8.0	0.96	0.065	1820	34.94	62.79	15.75	1.94	> 5000	(Solvay 2024)
PFPE	HT270	Solvay	270	14	25.9	0.96	0.065	1850	35.52	62.79	15.75	1.94	> 5000	(Solvay 2024)
PFO	Noah 3000A	Noah	110	1.35	2.5	1.159	0.061	1830	42.42		15.76	1.79	120	(Zhang et al., 2024)
PFO	Noah 3000D	Noah	112	1.353	2.5	1.177	0.062	1815	42.73					(Sun et al., 2025)
PFO	Noah 3000EP	Noah	124	1.38	2.5	n.d.	n.d.	1800	n.d.		> 14.28	2.09	108	(Zhejiang Noah Fluorochemical Co. Ltd 2025)
PFO	YL-70	Haohua	110–115	1.24	2.3	0.922	0.061	1830	33.75	70.10	13.46	2.09	1000–3000	(Solvay 2024)
PFC	ICELOONG 3123A	Yonghe	110	1.043	1.9	1.177	0.062	1812	42.65	92	18.8	1.91	120	(ZheJiang Yonghe Refrigerant Co. Ltd 2025)
	F Type		155	1.8	3.3	1.1	0.1	1850	40.70					(Wang et al., 2023)
	N Type		115	1.4	2.6	1.1768	0.1	1830	43.07		≥ 12.8			(Wang et al., 2023)
	C Type		> 175	3	5.4	0.9	0.07	1805	32.49		≥ 16	1.95		(Wang et al., 2024)
	X Type		165	2.2	4.1	1.1	0.065	1855	40.81		> 16	~1.9		(Wang et al., 2024)
	JX-1	Juhua		3.3	6.31	1.238	0.062	1888.9	46.77					(Ding et al., 2024)
	EFL-1	Juhua	177.3	1.5	2.77	1.165	0.068	1889	44.01					(Huang et al., 2024)
	EFL-2	Juhua	286.2	1.1	2.15	1.338	0.064	1894	50.68					(Huang et al., 2024)
	EFL-3	Juhua	172.5	0.8	1.49	1.219	0.059	1883	45.91					(Huang et al., 2024)
	SS110			0.7	1.18	1.055	0.063	1738	36.67					(Zhang et al., 2026)

Chang, 2022; Jin et al., 2024). Extensive studies on FC-40 (Khoshvaght-Aliabadi et al., 2026; Khoshvaght-Aliabadi et al., 2026) show that, the low viscosity of FC-40 facilitates efficient convective heat transfer, endowing it with superior performance compared to many coolants. Due to its lower viscosity than EC-130 and MO, it effectively reduces component temperatures (Vagiakis et al., 2025). Despite lower c_p and λ compared with PAO-6, the lower viscosity of FC-40 reduces flow resistance in heat sinks. Therefore, FC-40 exhibits superior cooling performance (Shrigondekar et al., 2023). Its viscosity further decreases with increasing thermal load, raising the Reynolds number (Re) and reducing thermal resistance (Muneeshwaran et al., 2023). As inlet temperature rises, viscosity and ρ decrease while c_p increases, enhancing heat transfer and reducing chip temperature under low-flow conditions (Wang et al., 2024). To evaluate the material compatibility, Liu et al. (Yu et al., 2022) found that FC-40 caused severe failure of phase-change thermal interface materials, which migrated into the coolant, indicating incompatibility. In contrast, the silicone-based material showed minimal weight change and stable thermal conductivity after long-term immersion, demonstrating good compatibility with FC-40.

Another widely studied PFAM coolant is FC-3283 (Liu et al., 2024; Liu et al., 2025). Compared with MO (Gajjar and Huang, 2023), SIO, and soybean oil (Matsuoka et al., 2017), it provides better cooling performance due to lower viscosity and thus lower T_a. Its signal integrity, reliability, and safety have also been evaluated. Kurata et al. (Kurata et al., 2021) demonstrated that its low dielectric constant reduces impedance along data paths and improves signal integrity. Min et al. (Min et al., 2025) further reported that, compared with air cooling, SPIC significantly reduces heat accumulation in DRAM packages under high-voltage stress and suppresses leakage current by about 59%. At voltages above 3.0 V, degradation accelerates while leakage current in air is over three times higher than in SPIC, confirming its electrical stability.

4.2. Hydrofluoroether

HFEs are obtained by introducing O atoms into hydrofluorocarbon (HFCs) or replacing F atoms with H atoms in PFPEs. By definition, the difference between HFEs and unsaturated hydrofluoroethers (UHFEs) lies in whether the compound is saturated. HFEs belong to hydrogenated saturated fluorocarbons. In data center SPICs, they are mainly represented by the Novec series from 3 M. HFEs exhibit good heat transfer performance, and their GWP is generally below 500, indicating a relatively lower greenhouse effect compared to other fluorocarbon coolants, with minimal ozone impact. Among them, Novec-7200, Novec-7300, and Novec-7500 have GWP below 250, making them environmentally friendly fluorocarbon coolants. Compared with PFAMs, HFEs have lower ρ, resulting in reduced q_v and weaker transient thermal buffering ability. However, due to stronger intermolecular interactions from their partially fluorinated ether structure, they exhibit higher vaporization latent heat. Their dielectric constant is typically above 5 because of molecular polarity and the presence of hydrogen atoms, which can increase electrical conductivity and affect signal integrity. Therefore, HFEs are more suitable for applications with less stringent dielectric requirements.

It is important to note that Novec-7100, Novec-7200, and Novec-7300 have boiling points below 100 °C and are generally unsuitable for SPIC systems, being mainly used as two-phase immersion coolants (Wang et al., 2023). However, due to their non-flammability, low toxicity, and zero ODP, Rokonuzzaman et al. (Rokonuzzaman et al., 2025) investigated their single-phase heat transfer performance under natural and forced convection, focusing on conditions below 54 W heat flux where Novec-7100 remains single-phase. Khoshvaght-Aliabadi et al. (Khoshvaght-Aliabadi et al., 2025) also reported SPIC applications of Novec-7200 and Novec-7300, noting that Novec-7200 has the lowest viscosity and highest thermal conductivity, resulting in better hydraulic performance.

Novec-7500 and Novec-7700 have boiling points above 120 °C, making them suitable for SPIC (Kim et al., 2025; Li et al., 2025; Khoshvaght-Aliabadi et al., 2025). Their low viscosity and high ρ provide superior heat transfer performance and lower ΔP. Based on the superior heat transfer performance and environmental friendliness of Novec-7500, Suresh et al. (Suresh et al., 2024) analyzed the thermal performance of a multicore server immersed in Novec-7500. However, compared to other coolants, it is relatively expensive.

4.3. Perfluoropolyether

PFPEs are obtained by introducing O atoms into PFCs or replacing all H atoms in HFEs with F atoms. PFPEs belong to the class of perfluorinated saturated compounds. PFPEs have low c_p, resulting in weak q_v and limited thermal buffering performance. Their weak intermolecular forces also lead to low vaporization latent heat, similar to PFAMs. Due to the low polarizability of F atoms (Liaw et al., 2012), PFPEs exhibit very low dielectric constants (below 2) and high dielectric strength (up to 40 kV at a 2.54 mm gap), making them suitable as immersion coolants from a technical standpoint. However, their GWP exceed 5000, indicating strong greenhouse effect, and thus limiting their environmental sustainability.

Currently, PFPEs used in SPIC systems are mainly the HT series from Solvay (Solvay, 2024). The physical and chemical property data available are sourced from Galden® HT specifications. Their boiling points range from 110 to 270 °C, the highest among fluorocarbon coolants used in SPICs. However, they are relatively expensive. Furthermore, although PFPEs possess high thermal stability and chemical inertness, their non-environmentally friendly and non-biodegradable properties pose risks to the environment and human health. Therefore, the production and use of PFPEs should also be restricted or even banned.

4.4. Perfluoroolefin

PFOs are obtained by introducing C═C double bonds into PFCs. They can also be produced by replacing all H atoms in hydrofluoroolefin (HFOs) with F atoms. PFOs are a class of perfluorinated unsaturated compounds. They offer excellent cooling performance, good material compatibility, low toxicity, and ease of synthesis. They deliver favorable sensible heat performance due to their proper ρ and thermal property characteristics. PFOs have a low dielectric constant and high dielectric strength, resulting in excellent insulating properties. Furthermore, their GWP is low (except for YL-70, which has a GWP $\leq$ 3000). The GWP of the Noah series of PFOs developed by Noah Company in Zhejiang, China, is below 150 (Zhejiang Noah Fluorochemical Co. Ltd, 2025). The GWP of PFOs is significantly lower than that of PFAMs and PFPEs, essentially meeting the environmental standards (except for YL-70). A comparison of the environmental performance and molecular structures of these three compounds reveals that the low GWP of PFOs is primarily attributable to the presence of internal unsaturated bonds, which enable rapid degradation in the atmosphere. Therefore, PFOs represent a data center SPIC coolant with significant application potential.

Some researchers have already conducted comparative studies on PFOs. Due to the low viscosity of Noah 3000A, its relative turbulent flow characteristics cause its thermal resistance to decrease much more significantly than that of PAO-4 as flow rate changes. Under high flow rate conditions, the thermal resistance of Noah 3000A is slightly lower than that of PAO-4 (Zhang et al., 2024). Compared to natural MO, SIO, and SYO ILC-5, the high ρ and low viscosity of the Noah 3000D result in the lowest coolant temperature and smallest temperature difference in SPIC systems (Sun and Liu, 2026). The superior performance and acceptable cost of PFOs will further facilitate their broader application in data center SPIC systems. Currently, many companies are actively promoting their adoption (Zhang et al., 2022).

5. Nanofluid coolants

In addition to the two most commonly used types of immersion coolants above, nanofluid coolants, an emerging class of modified coolants, have attracted widespread attention from researchers in the field of data center SPIC. Due to their unique enhanced heat transfer properties, they offer a potential pathway to overcome the thermal bottlenecks of high-power density chips (Sohel Murshed and Nieto de Castro, 2017). Nanofluids can be broadly categorized into oil-based nanofluids and water-based nanofluids.

5.1. Oil-based nanofluids

Oil-based nanofluids are formed by dispersing highly thermally conductive nanoparticles into a base fluid such as MO (Rehman et al., 2026). Their thermophysical properties are jointly determined by the base fluid and nanoparticles. The base fluid governs the fundamental characteristics, while nanoparticles enhance heat transfer through their high thermal conductivity, micro-scale effects, and Brownian motion. Meanwhile, nanoparticle size, shape, concentration, and dispersion state significantly influence viscosity and flow behavior. MO is commonly used as the base fluid due to its low cost, wide availability, mature processing technology, and suitability for large-scale preparation. It also has moderate viscosity, which facilitates nanoparticle dispersion, and good chemical stability, ensuring compatibility with additives. Thus, MO provides a balanced platform between performance and cost.

Common nanoparticles and their thermophysical properties are shown in Table 6. Current studies on oil-based nanofluids include SiC (Li et al., 2016), Al_2O_3 (Umar et al., 2020), hexagonal boron nitride (HBN) (Thomas et al., 2012), SiO_2 (Luo et al., 2023), and composite (SiC/TiO_2) oil-based nanofluids (Wei et al., 2017).

The type, concentration, and shape of nanoparticles have varying effects on thermophysical properties. For SiC nanofluids, Luo et al. (Luo et al., 2022) reported that a 0.3% volume fraction performs better at low Re, while 3.7% is more effective at high Re. For Al_2O_3 nanofluids, higher concentrations improve heat transfer and reduce maximum CPU temperature, but increase viscosity and ΔP (Niazmand et al., 2020). For CuO nanofluids, Sristi et al. (Sristi et al., 2025) showed that adding CuO to olive oil significantly enhances thermophysical properties, increasing thermal conductivity by 15.29%, while thermal diffusivity improves and specific heat decreases with concentration. For composite nanofluids, Shelton et al. (Shelton et al., 2023) demonstrated that Al_2O_3-TiO_2 nanofluids behave more closely to Newtonian fluids than single-component nanofluids. Similarly, Asifa et al. (Asifa et al., 2025) found that adding MoS_2 and Al_2O_3 in equal proportions to MO can increase λ by nearly 67%. The nanofluid exhibited optimal thermal performance when nanoparticles are in a lamina form, accompanied by an 11.2% reduction in flow resistance.

Table 6
Nanoparticles and properties.

Name	Specific heat capacity c_p/ $kJ\cdot kg^{-1}\cdot K^{-1}$	Thermal conductivity $\lambda/W\cdot m^{-1}\cdot K^{-1}$	Density $\rho/kg\cdot m^{-3}$	Ref
SiC	0.422	165	1100*	(Luo et al., 2022; Li et al., 2016)
Al_2O_3	0.765	40	3970	(Niazmand et al., 2020)
SiO_2	0.966	4	2200	(Luo et al., 2023)
HBN	0.71	20–30	2290	(Luo et al., 2023)
Alu	0.96	173	2810	(Asifa et al., 2025)
MoS_2	0.397	904	5060	(Asifa et al., 2025)
Cu	0.385	401	8933	(Manzoor et al., 2026)
CuO	0.55	20	6300	(Alkasmoul et al., 2025)
TiO_2	0.72	12	4200	(Alkasmoul et al., 2025)

Note: The density of SiC shown in the table is the bulk density of SiC nanoparticles.

Further investigations have examined the long-term stability and dielectric properties of nanofluids. Koutras et al. (Koutras et al., 2022) investigated TiO_2 and SiC nanofluids in NE at different concentrations and found that thermal aging can improve dielectric performance. SiC nanofluids showed better high-temperature stability than TiO_2, with the best dielectric response and aging stability observed at a mass fraction of 0.004%. Mai et al. (Mai et al., 2025) modified Al_2O_3 nanoparticles using hexadecyl trimethoxy silane (HDTMS), significantly improving the stability of MO-based nanofluids. The 0.4 wt% modified sample showed the best performance due to enhanced nanoparticle–fluid interactions that suppressed aggregation.

Furthermore, stronger interparticle interactions and improved nanoparticle-fluid coupling can reduce internal energy fluctuations and enhance thermophysical properties. Doping with nanoparticles such as SiO_2 can also regulate volumetric heat capacity and improve transient thermal buffering performance (An et al., 2026). However, research on the thermal buffering mechanisms of oil-based nanofluids remains limited, and the synergistic effects of nanoparticles on heat buffering behavior require further systematic investigation.

5.2. Water-based nanofluids

In air-cooled data center systems, Al_2O_3 water-based nanofluids can reduce CPU temperatures by up to 20% compared with conventional coolants. TiO_2 and CuO nanofluids also show similar improvements in thermal efficiency, particularly in CPU and microchip heat sink applications (Rafati et al., 2012; Nazari et al., 2014; Ghasemi et al., 2021).

In SPIC systems, the applicability of water-based nanofluids has also been investigated. Alkasmoul et al. (Alkasmoul et al., 2025) compared Al_2O_3, TiO_2, and CuO water-based nanofluids and found that Al_2O_3 exhibits the best thermal performance due to its higher thermal conductivity, significantly reducing T_a and improving energy efficiency, However, its higher viscosity leads to increased ΔP and pumping costs. Makaoui et al. (Makaoui et al., 2025) found that, for Cu-water nanofluids, higher Re and Rayleigh numbers (Ra) enhances convective and buoyancy-driven heat transfer. As Re and nanoparticle concentration increase from 0% to 5%, heat transfer can improve by up to 193.8%, while combined increases in Ra and concentration yield enhancements of up to 36.3%. Since water is electrically conductive, insulating coatings are required for safe operation in SPIC systems. Therefore, water-based nanofluids can also be considered within the broader category of conductive fluid coolants discussed in Section 6.

At present, nanofluids are still largely at the research and development stage, with only limited pilot-scale applications. Their relatively high preparation cost and several practical challenges hinder large-scale deployment. A key issue is stability: nanoparticles tend to agglomerate, forming clusters that reduce λ and increase viscosity (Rahman et al., 2024), eventually leading to sedimentation and separation from the base fluid. For this reason, the optimal particle concentration is generally kept within 0–5%. Higher loadings increase the risk of instability, deposition, and flow blockage, which can degrade heat transfer performance and system reliability over time. In addition, nanoparticle-induced erosion of electronic components during circulation and changes in electrical conductivity and dielectric properties may compromise insulation safety.

Despite these challenges, research has proposed various improvement strategies, particularly nanoparticle surface modification and the development of effective dispersants to enhance stability and dispersion. With continued advances in material engineering and formulation design, nanofluids are expected to remain a promising candidate for high-performance data center cooling and may play an important role in the future development of advanced liquid cooling technologies.

6. Conductive fluid coolants

For electrical safety reasons, the SPIC coolants mentioned above are dielectric fluids. However, compared to high-performance conductive fluids such as water, dielectric fluids have relatively inferior thermophysical properties (Birbarah et al., 2020). Conductive fluids offer superior heat transfer performance, making them attractive candidates for cooling high power density electronic devices. They also offer higher c_p than dielectric fluids, providing greater transient thermal buffering potential. Among them, water is the most accessible conductive fluid, with excellent heat capacity, heat transfer performance, and low cost. In addition to conventional hot-water thermal storage, water immersion cooling (WIC) has therefore attracted increasing attention (Li et al., 2022).

The main barrier to WIC is the electrical conductivity of water. This issue can be mitigated by applying insulating coatings to electronic components, enabling high heat transfer while maintaining electrical safety. Koibuchi et al. (Koibuchi et al., 2019) developed a parylene-coated water-immersed computer and achieved a 20 °C reduction in chip temperature, while also increasing operating frequency. Similarly, Gebrael et al. (Gebrael et al., 2022) used parylene coatings and integrated Cu layers directly on devices, eliminating thermal interface materials and reducing thermal resistance, thereby improving temperature stability and cooling performance.

Currently, research on conductive fluid cooling in SPIC systems remains limited due to safety concerns associated with electrical conductivity. Any damage to insulating coatings caused by wear or thermal aging may lead to leakage and potential short-circuit risks. Therefore, conductive fluid cooling requires highly reliable coating materials and manufacturing processes. It is currently more suitable for small-scale data centers. Although it offers heat transfer advantages, its engineering complexity and safety risks make it unlikely to replace dielectric coolants as the mainstream solution in large-scale SPIC systems in the near future.

7. Comparative assessment of immersion coolants

Comparing the performance differences among various coolants and systematically evaluating them based on a series of key metrics is a critical step in selecting an appropriate coolant and optimizing cooling system design. The performance characteristics of the oil and fluorocarbon coolants mentioned above have been summarized, and the results are presented in Table 7.

First, a comparative analysis of the thermophysical properties is essential for performance evaluation and selection. As shown in Table 7, oil coolants have higher c_p and λ, while fluorocarbon coolants have lower ν, higher ρ and higher q_v. Khoshvaght-Aliabadi et al. (Khoshvaght-Aliabadi et al., 2025) found that among the tested coolants, Novec-7200 exhibits the best overall performance due to its excellent thermophysical properties. Furthermore, benefiting from the inherently high c_p, oil-based coolants can store more heat per unit mass, providing strong mass-based sensible heat and effective thermal buffering. By contrast, fluorinated liquids are restricted by low c_p in mass-scale thermal buffering, while their high ρ greatly improves q_v, achieving better volume-based thermal buffering. The combination of low ν and high q_v makes fluorocarbon coolants particularly suitable for high power density servers, where efficient convective heat transfer and rapid transient thermal response are essential under fluctuating AI workloads.

Given the interdependence of thermophysical properties, the Figure of Merit (FOM) has been introduced to provide a more integrated evaluation of coolant performance. For immersion coolants, various

Table 7
Comparison of coolant properties.

Property	oil coolants	Fluorocarbon coolants
Kinematic viscosity $\nu/\mathrm{mm^2{\cdot}s^{-1}}$	$\approx > 5$	< 5 (Besides FC-70 and HT270)
Specific heat capacity c_p/ $\mathrm{kJ{\cdot}kg^{-1}{\cdot}K^{-1}}$	> 1.45	< 1.35
Thermal conductivity $\lambda/\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$	≥ 0.11	≤ 0.1
Density $\rho/\mathrm{kg{\cdot}m^{-3}}$	< 1100	> 1400
Volumetric-specific sensible heat q_v/ $\mathrm{MJ{\cdot}m^{-3}}$ ($\Delta T = 20$ °C)	28–43	32–51
Dielectric strength $/\mathrm{kV{\cdot}mm^{-1}}$	$\approx \geq 20$	> 9.84
Dielectric constant	$\approx < 2.25$ Some (SYOs) >3.3 SIOs: 2.6–2.71	< 2.1 (Besides HFEs) > 5 (HFEs)
GWP	0	> 5000 (PFAMs and PFPEs) < 250 (HFEs besides Novec-7100 and Novec-7700) < 150 (PFOs besides YL-70)

FOM formulations have been proposed under different operating conditions, as shown in Table 8. The FOM combines ρ, λ, c_p, and viscosity, and is used to assess coolant heat transfer capability and its impact on IT equipment performance. It should be noted that these FOM equations are derived under specific flow regimes and driving mechanisms. Equations for natural convection emphasize buoyancy-driven flow and may deemphasize thermal conductivity. Meanwhile, equations for forced convection place varying weights on ρ, λ, c_p, and viscosity depending on whether the flow is laminar or turbulent, and whether pressure drop is a primary concern. Therefore, the selection of an appropriate FOM equation should be guided by the actual operating conditions of the cooling system, and direct comparison of FOM values across different formulations should be avoided.

Among the forced convection equations, No (5) and (6) are considered to be more universally applicable within their respective flow regimes. Therefore, these two equations serve as suitable bases for comparing the cooling performance of different immersion coolants. They provide a balanced and theoretically consistent evaluation across a wide range of thermophysical properties. For instance, compared to SL 3568, BioLife 4 shows a twofold increase in natural convection FOM (FOM1) and a tenfold reduction in Prandtl number, leading to improved thermal performance and about a 6 °C reduction in component temperature (Ruichek et al., 2025). A higher FOM generally indicates better heat transfer performance. Among various fluorocarbon and oil coolants, SF-10 exhibits the highest FOM, ranging from 17.2% to 50.6% higher than others, indicating superior thermal performance (Li et al., 2025). Furthermore, Chen et al. (Chen et al., 2024) conducted thermal property tests and FOM comparative analyses, establishing the following ranking of thermal properties for three electronic fluorinated liquids (EFLs): $\mu > \lambda > \rho > c_p$. For the SPIC cabinet using EFL with the lowest μ, Nusselt number increases by 57.3%, and ΔP decreases by 59%.

In addition to thermal performance, coolants must satisfy requirements in material compatibility, signal integrity, electrical insulation, toxicity, and environmental impact. Based on Table 7 and the comparison above, oil coolants generally exhibit high dielectric strength, low dielectric constant, and low GWP (Li et al., 2025). Fluorocarbon coolants offer good material compatibility, high operational reliability, and low dielectric constants (except HFEs), but their dielectric strength is relatively lower and their GWP is high. HFEs, in particular, may affect signal compatibility due to higher dielectric constants.

Meanwhile, it is undeniable that the greenhouse effect and environmental persistence associated with fluorocarbon coolants, many of which fall under the category of per- and polyfluoroalkyl substances (PFASs), pose significant environmental concerns. In response to these

Table 8
FOM associated equations.

Number	Equation	Condition	Limitation	Applicability	Ref
1	$FOM = \frac{\rho\beta c_p}{\mu}$	Laminar flow & natural convection	Ignore λ	Thermal drive is far more important than thermal conductivity	(Singha et al., 2017)
2	$FOM = \lambda\left(\frac{\beta c_p\rho^2}{\mu\lambda}\right)^{0.2813}$	Laminar flow & natural convection	Comprehensive consideration	More universal	(Van Vaerenbergh et al., 2025; Ruichek et al., 2025)
3	$FOM = \frac{\lambda c_p}{\mu}$	Laminar flow & forced convection	Ignore ρ	Density and volumetric flow rate have little effect Suitable for high-viscosity fluids	(Kelly et al., 2021)
4	$FOM = \lambda\left(\frac{c_p\rho}{\lambda}\right)^{1/3}$	Laminar flow & forced convection	Ignore μ	Suitable for low-viscosity fluids	(Van Vaerenbergh et al., 2025; Open Compute Project 2022)
5	$FOM = \left(\frac{\rho {c_p}^2\lambda^2}{\nu}\right)^{0.25}$	Laminar flow & forced convection	Comprehensive consideration	More universal	(Ehrenpreis et al., 2020)
6	$FOM = \frac{\lambda^{0.6}{c_p}^{0.4}\rho^{0.8}}{\mu^{0.4}}$	Turbulent flow & forced convection	Comprehensive consideration	Derivation based on the Dittus-Boelter equation; More universal	(Li et al., 2025; Chen et al., 2024)
7	$FOM = \frac{\lambda^{0.67}{c_p}^{1.33}\rho^{0.05}}{\mu^{0.72}}$	Turbulent flow & forced convection	Comprehensive consideration, but almost ignoring ρ	Derivation based on the Gnielinski equation; Density difference is small	(Kelly et al., 2021)

growing pressures, regulatory bodies worldwide have begun to take action. The European Union, for instance, has initiated legislative measures to prohibit the use of PFASs in certain products in the near future (European Chemicals Agency, 2024). Companies such as 3 M have announced the discontinuation of PFASs-related product development and manufacturing. Regulatory agencies in other regions, including Canada, Australia, and Asia, have also begun to take note of the risks posed by PFASs. These developments underscore a broader industry trend: future SPIC technologies will place greater emphasis on the research and development of environmentally friendly coolant alternatives.

Furthermore, as for emerging coolant technologies, nanofluid coolants can enhance heat transfer, but challenges such as dispersion stability, sedimentation and agglomeration, and pipeline wear remain to be addressed. While ensuring equipment safety, conductive fluid coolants combined with insulating coatings can substantially improve heat transfer performance. However, the coating durability, long-term reliability, and preparation processes are still under investigation.

The above analysis shows that no single coolant can simultaneously achieve optimal performance across all key indicators. Different coolants therefore exhibit varying suitability depending on requirements such as power density, operational demands, and environmental constraints. The comparative results are summarized in Table 9. It should be noted that nanofluid and conductive fluid coolants are not included in this comparative rating, as their thermophysical properties and material compatibility data are not yet sufficiently established in the literature.

Table 9
Comparison of immersion coolant adaptability across different scenarios.

Coolant	Subcategory	Heat dissipation performance (FOM×10^3 No.6)	Transient thermal buffering performance (q_v/MJ·m^{-3})	Signal compatibility (Dielectric constant)	Insulation performance (Dielectric strength/kV·mm^{-1})	Material compatibility	Economy (Cost)	Environmental performance (GWP combined with biodegradability)	Adaptability of power density scenarios (FOM×10^3 No.6)
Oil coolants	MO	6.60-12.49	28.36-40.00	~ 2.05	31.2	Limited compatibility; Prone to fouling and clogging	Extremely low cost	0	6.60-12.49
	SYO	4.53-18.28	31.35-38.70	2.02-2.21 (Besides L759 and Therminol VP-1)	16.8-30.8	Good to moderate compatibility; Label adhesive incompatibility	Low cost	0	4.53-18.28
	SIO	3.61-10.67	28.02-30.72	2.60-2.71	25-27.5	Moderate compatibility; Hydrolysis and oxidative deposition	Low cost	0	3.61-10.67
	NE	5.23-7.30	36.77-42.55	3.1-3.3	45-55	Moderate compatibility	Relatively high cost	GWP = 0; Easily degradable	5.23-7.30
Fluorocarbon coolants	PFAM	6.41-18.89	37.22-42.68	1.80-1.98	15.75-18.1	Good compatibility	High cost	> 5000	6.41-18.89
	HFE	12.52-23.58	33.56-39.01	5.48-7.40	9.84-13.78	Good compatibility	High cost	2.5-420	12.52-23.58
	PFPE	5.36-16.57	32.83-35.52	1.92-1.94	15.75	Good compatibility	High cost	> 5000	5.36-16.57
	PFO	13.33-14.28	33.75-42.73	1.79-2.09	13.46-15.76	Good compatibility	High cost	108-120 (Besides YL-70)	13.33-14.28

Note: Heat dissipation performance is characterized by FOM No. (6) in Table 8. Darker shading (e.g.,) indicates superior performance, and lighter shading (e.g.,) indicates inferior performance.

Moreover, they remain largely at the laboratory research stage, with challenges such as dispersion stability, sedimentation, agglomeration, wear, and coating durability still unresolved. Nevertheless, they are regarded as having considerable potential as next-generation coolants for ultra-high power density data centers. This is attributed to their exceptional heat transfer potential and the ability to combine high thermal conductivity with electrical insulation through protective coatings. Their inclusion in future comparative frameworks is anticipated as more comprehensive data become available.

Based on the comparison in Table 9, the following application-oriented recommendations can be made. For cost-sensitive applications, MOs are preferred due to their significant cost advantage, which helps reduce data center construction expenses. SYOs offer a balanced trade-off between cost, thermal performance, and environmental impact, making them a versatile choice for general-purpose applications. Under strict environmental requirements, NEs are more suitable because of their superior environmental performance. Under variable heat load conditions, NEs, PFAMs and partial PFOs exhibit prominent transient thermal buffering performance owing to their high ρ and excellent q_v, which facilitates effective temperature control within compact cabinet layouts. For high power density applications requiring excellent heat transfer performance and long-term reliability, fluorocarbon coolants are currently the most widely adopted option, despite their higher cost and environmental concerns.

Overall, a multi-dimensional evaluation of key performance metrics is essential for matching coolants with specific application scenarios. Meanwhile, the development of an integrated performance index that synthesizes these key metrics into a holistic evaluation framework is equally important for enabling systematic comparison across different coolants. Moreover, the insights derived from such evaluations can ultimately guide the optimization of existing coolants and the development of novel coolants with balanced performance across multiple dimensions, thereby providing guidelines for the selection, improvement, and future design of immersion coolants.

8. Opportunities for future research

As data centers evolve toward higher power density and greener, low-carbon operation, the widespread adoption of SPIC technology imposes increasingly stringent requirements on coolant performance. Although existing coolants offer distinct technical advantages, they also exhibit limitations that cannot be ignored. Future optimization should aim to overcome performance bottlenecks while preserving their inherent benefits, enabling reliable long-term operation under high heat flux conditions. Key directions are summarized as follows:

- **Improve reliability of oil coolants:** Oil coolants, especially MOs, suffer from limited long-term stability. The use of composite additives (Open Compute Project, 2022) can enhance oxidation resistance, corrosion resistance, and overall stability, extending service life. For incompatible materials such as certain adhesives (Beau et al., 2024; Shah et al., 2021), alternative or newly developed compatible materials are needed. In addition, establishing a material compatibility database and standards for data centers can reduce repeated verification efforts.
- **Reduce cost and GWP of fluorocarbon coolants:** Given the high costs and GWP of fluorocarbon coolants, optimizing the synthesis process (Zhang et al., 2022) and developing cleaner, simpler production routes can reduce both cost and emissions. Molecular design strategies, such as introducing C=C bonds, can help lower GWP. Furthermore, recovery, purification, and recycling systems are essential to enable resource reuse and reduce lifecycle cost.
- **Enhance stability and lower cost of nanofluids:** To mitigate particle agglomeration and sedimentation, stability control mechanisms and optimized dispersion techniques (ultrasonic treatment and surface modification) are required (Mai et al., 2025; Javidi and Entezari, 2024). Scalable low-cost preparation methods should also be prioritized. Moreover, due to the lack of standardized evaluation systems, unified metrics balancing thermal performance and electrical safety are needed to support engineering adoption.
- **Optimize insulating coating technology for conductive fluids:** Conductive fluids require reliable insulating coatings to ensure safe operation. Research should focus on improving coating insulation strength, adhesion, and durability (Zhu et al., 2026). Optimizing coating materials and processes, as well as reducing defects such as pinholes and cracks, will improve compatibility with fluids and substrates and ensure long-term operational safety.
- **Develop blended modified composite coolants:** Mixing different coolants can combine complementary advantages and overcome individual limitations. For example, NE-synthetic ester blends (Qian et al., 2025) improve both environmental performance and thermal capability, while oil-fluorocarbon blends (Fang et al., 2025) balance cost and heat transfer performance. Further formulation optimization is needed to develop high-specific heat, high-density, and high-stability composite coolants that meet both cooling and thermal buffering requirements.
- **Establish a comprehensive evaluation framework for coolants:** Future research should develop a unified performance index that integrates heat dissipation, thermal buffering, signal integrity, insulation, compatibility, cost, and environmental impact into a holistic framework, enabling objective comparison and clear guidance for coolant selection and optimization.
- **Address practical challenges for large-scale deployment:** While the aforementioned directions provide solutions for coolant-specific performance and reliability issues, integrating these solutions into a coherent framework for large-scale deployment remains a critical challenge. Future efforts should focus on long-term reliability, maintenance protocols, and cost-effective recycling strategies to ensure sustainable deployment of SPIC systems.

These research directions collectively provide a comprehensive roadmap for the development of next-generation coolants, spanning from molecular design and material optimization to system-level integration and lifecycle management, ultimately facilitating the widespread deployment of SPIC technology in high-performance, energy-efficient data centers.

9. Conclusions

Existing reviews of SPIC technology focus primarily on the systems. The coverage of coolants remains brief and fragmented, with oversimplified classifications, insufficient performance analysis, and inadequate attention to emerging types. This work presents a dedicated coolant-centered review. A refined classification system has been established, covering both traditional dielectric coolants and new functional coolants. The transient thermal buffering performance evaluation under dynamic loads is introduced. Multi-dimensional comparisons of thermophysical, electrical, economic, and environmental properties are conducted. Future development directions and challenges are also discussed. This comprehensive framework provides clearer guidance for coolant research, selection, and large-scale application. The main conclusions are as follows:

- Oil coolants are low-cost, widely available, and provide good electrical insulation, making them suitable for medium-to-low power density scenarios. Their high c_p also gives strong mass-based thermal buffering potential. However, they suffer from limited cooling performance, poor long-term reliability, and compatibility issues. Improvements require additives, blending strategies, and compatible material development.
- Fluorocarbon coolants offer excellent heat transfer performance, chemical stability, and reliability, making them suitable for high

power density applications. Their high density provides strong volumetric thermal buffering capacity. However, they are expensive, have high GWP, and complex synthesis processes. Future work should focus on low-GWP alternatives, process optimization, recycling, and blending with oils to reduce cost and carbon footprint.
- Nanofluids provide high heat transfer capability for ultra-high power density cooling but face challenges such as instability, particle agglomeration, lack of standards, and high cost. They remain primarily at the research stage, and significant advances in preparation processes, stability control, and reliability validation are required before their engineering application can be realistically considered.
- Conductive fluids offer high heat transfer efficiency and strong environmental performance, but their intrinsic electrical conductivity precludes direct use in immersion cooling without protective insulation. With their long-term durability still unproven and substantial material and safety validation still required, these fluids are not yet suitable for commercial deployment in data center thermal management.
- Beyond improving the individual performance of each coolant type, a unified comprehensive performance index should be established. This index can integrate multiple properties into a holistic framework for objective comparison. More importantly, systematic efforts are required to advance long-term reliability validation, maintenance protocols, and cost-effective recycling strategies, thereby ensuring the sustainable large-scale deployment of SPIC systems.

In the future, SPIC coolant research will continue to develop toward higher efficiency, lower environmental impact, lower cost, and longer service life. Mature technologies are expected to enable widespread use of balanced solutions combining cooling and thermal buffering in high-performance data centers. Beyond data centers, the findings on coolant performance, compatibility, and evaluation methods can also inform the development of dielectric fluids for other high heat flux applications, supporting broader progress toward green and efficient thermal management technologies.

CRediT authorship contribution statement

Yaran Liang: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Shuangquan Shao:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Key R&D Program for Technological Innovation of Hubei Province (No.2025BAB003) and National Natural Science Foundation of China (No.52576012).

References

Acton, M., Booth, J., Paci, D., 2025. 2025 Best Practice Guidelines For the EU Code of Conduct on Data Centre Energy Efficiency. Publications Office of the European Union, Luxembourg.

Agung Pambudi, N., Muhamad Yusuf, A., Sarifudin, A., 2022. The use of single-phase immersion cooling by using two types of dielectric fluid for. Data Cent. Energy Sav. Energy Eng. 119 (1), 275–286.

Al Kez, D., Foley, A M, Hasan Wong, F W B M, Dolfi, A., Srinivasan, G., 2025. AI-driven cooling technologies for high-performance data centres: state-of-the-art review and future directions. Sustain. Energy Technol. Assess. 82, 104511.

Alkasmoul, F S, Almogbel, A M, Shahzad, M W, Al-damook, A J, 2025. Thermal performance and optimum concentration of different nanofluids in immersion cooling in data center servers. Results. Eng. 25, 103699.

AMD, 2026. Trend chart of TDP. www.amd.com [accessed 25 March 2026].

An, Z., Mao, S., Du, X., Zhang, D., Fu, J., Ding, Y., 2026. In-depth molecular dynamics analysis of the thermal energy storage and transfer enhancement mechanism within carbonate nanofluid system. Int. J. Heat. Mass Transf. 255, 127825.

ARES-WHI0, 2026. Intel whitely. https://www.aaeon.com/vi/product/detail/intel-ice-lake-xeon-server-board-ares-whi0/specification [accessed 21 July 2026].

Asifa, Anwar T, Kumam, P., Almutairi, K S, Watthayu, W., 2025. Thermophysical dynamics of mineral oil-based hybrid nanofluids under multiple flow conditions and radiation effects; individual, synergistic, and shape impacts analysis. J. Radiat. Res. Appl. Sci. 18 (4), 101748.

Azarifar, M., Arik, M., Chang, J-Y., 2024. Liquid cooling of data centers: a necessity facing challenges. Appl. Therm. Eng. 247, 123112.

Bansode, P., Suthar, R., Bhandari, R., Lakshminarayana, A., Eda, N T, Gupta, G., Simon, V., Modi, H., Nair, V., Shahi, P., Saini, S., Sivaraju, K B, Agonafer, D., 2024. Impact of immersion cooling on thermomechanical properties of halogen-free substrate core. J. Electron. Packag. 146 (4).

Beau, V V, Marion, K., Pieter, S., 2024. Performance evaluation of a next generation Ester based dielectric for single-phase precision immersion cooling. In: 2024 IEEE 5th International Conference on Dielectrics (ICD).

Bhandari, R., Lakshminarayana, A B, Sivaraju, K B, Bansode, P V, Kejela, E., Agonafer, D., 2022. Impact of immersion cooling on thermomechanical properties of non-halogenated substrate. In: ASME 2022 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.

Birbarah, P., Gebrael, T., Foulkes, T., Stillwell, A., Moore, A., Pilawa-Podgurski, R., Miljkovic, N., 2020. Water immersion cooling of high power density electronics. Int. J. Heat. Mass Transf. 147, 118918.

Bulinski M J, Tuma P E, Costello M G, Lamanna W M, Smith S M. Fluids for immersion cooling: US, US20200178414A1. 2020-06-04.

Cai, S., Gou, Z., 2026. Towards energy-efficient data centers: a comprehensive review of passive and active cooling strategies. Energy Built Environ. 7 (1), 206–226.

Chauhan, T., Bhandari, R., Sivaraju, K B, Chowdhury, A S M R, Agonafer, D., 2021. Impact of immersion cooling on thermomechanical properties of low-loss material printed circuit boards. J. Enhanc. Heat Transf. 28 (7), 73–90.

Chen, X., Huang, Y., Xu, S., Bao, C., Zhong, Y., Chen, Y., Zhang, C., 2024. Thermal performance evaluation of electronic fluorinated liquids for single-phase immersion liquid cooling. Int. J. Heat. Mass Transf. 220, 124951.

Cheng, C-C, Chang, P-C, Li, H-C, Hsu, F-I., 2020. Design of a single-phase immersion cooling system through experimental and numerical analysis. Int. J. Heat. Mass Transf. 160, 120203.

China Academy of Information and Communications Technology. Blue book on advanced computing and computing power development index (2025). 2025.

Das, S K, Choi, S U S, Patel, H E, 2006. Heat transfer in nanofluids—A review. Heat Transf. Eng. 27 (10), 3–19.

Ding, B., Xia, F., He, L., Huang, Y., 2024. A novel concept for performance enhancement of immersion-cooled data center servers. Int. Commun. Heat Mass Transf. 159, 107980.

Ding, J., Jia, Q., Chen, N., Li, Z., Zhang, J., 2025. Effect of droping with different proportions of mineral oil on performance of natural ester insulating oil. Insul. Mater. 58 (8), 67–74.

Ehrenpreis, C., Bahi, H E, Xu, H., Roux, G., Kneer, R., Rohlfs, W., 2020. Physically-motivated figure of Merit (FOM) assessing the cooling performance of fluids suitable for the direct cooling of electrical components. In: 2020 19th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems.

Eiland, R., Edward Fernandes, J., Vallejo, M., Siddarth, A., Agonafer, D., Mulay, V., 2017. Thermal performance and efficiency of a mineral oil immersed server over varied environmental operating conditions. J. Electron. Packag. 139 (4).

Eiland, R., Fernandes, J., Vallejo, M., Agonafer, D., Mulay, V., 2014. Flow rate and inlet temperature considerations for direct immersion of a single server in mineral oil. In: 14th IEEE ITHERM Conference.

European Chemicals Agency, 2024. Per- and polyfluoroalkyl substances (PFAS). https://echa.europa.eu/hot-topics/perfluoroalkyl-chemicals-pfas [accessed 4 March 2026].

Fan, C., Yang, R., Guo, H., Jiang, H., Zhang, C., Chen, Y., 2025. Efficient immersion cooling for electronic devices based on multi-physics field coupling. Int. J. Therm. Sci. 215, 110005.

Fang L, Yan T, Lu L. A composite immersion coolant, preparation method therefor, and application thereof: china, CN121108951A. 2025-12-12.

Finance News Network, 2025. TotalEnergies: pioneering green cooling innovation, empowering data centers' Low-carbon transition. https://www.chinaidr.com/tradenews/2025-10/251004.html [accessed 2 March 2026].

Fu, L., Fei, X., Guan, X., Cen, S., Ouyang, H., Guo, Z., 2025. Application research advances of fluorinated coolants for immersion liquid cooling in data center. Refrig. Air-Cond. 25 (6), 68–75.

Gajjar, K., Huang, H-P., 2023. Conjugate heat transfer for single phase immersion cooling of CPU. Case Stud. Therm. Eng. 52, 103728.

Gebrael, T., Li, J., Gamboa, A R, Ma, J., Schaadt, J., Horowitz, L., Pilawa-Podgurski, R., Miljkovic, N., 2022. High-efficiency cooling via the monolithic integration of copper on electronic devices. Nat. Electron. 5 (6), 394–402.

Ghasemi, S E, Ranjbar, A A, Hoseini, M J, Mohsenian, S., 2021. Design optimization and experimental investigation of CPU heat sink cooled by alumina-water nanofluid. J. Mater. Res. Technol. 15, 2276–2286.

Guo, L., Wu, J., Hsieh, I M, Huang, C-H, Li, B., Jao, C., Lin, V., Yuan, R., Tan, X., Fan, X., Wu, Z., Sheng, Y., He, Y., Zhang, J., Zhang, J., Chen, C., Xia, Y., Lv, W., Wu, M., Du, C., He, C., Zhang, J., Ahuja, N., Qiao, Q., 2021. Advanced single-phase passive immersion cooling solution with natural convection for outdoor edge servers. In: 2021 20th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm).

Han, X., Wang, Q., Zheng, J., 2016. Determination and evaluation of the optical properties of dielectric liquids for concentrating photovoltaic immersion cooling applications. Solar Energy 133, 476–484.
Hnayno, M., Chehade, A., Klaba, H., Polidori, G., Maalouf, C., 2023. Experimental investigation of a data-centre cooling system using a new single-phase immersion/liquid technique. Case Studies in Thermal Engineering 45, 102925.
Hu, Y., David, M.P., 2024. Immersion oil thermal performance characteristics and comparisons. In: 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm).
Huang, Y., Deng, Z., Chen, Y., Zhang, C., 2023a. Performance investigation of a biomimetic latent heat thermal energy storage device for waste heat recovery in data centers. Appl. Energy 335, 120745.
Huang, Y., Ge, J., Chen, Y., Zhang, C., 2023b. Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers. Int. J. Heat. Mass Transf. 207, 124023.
Huang, Y., Liu, B., Xu, S., Bao, C., Zhong, Y., Zhang, C., 2024. Experimental study on the immersion liquid cooling performance of high-power data center servers. Energy 297, 131195.
Huang, Y., Liu, C., Ding, B., Deng, Z., Gao, W., 2025. Machine learning-optimized jet-enhanced immersion liquid cooling for high-power data centers. Int. Commun. Heat Mass Transf. 169, 109654.
Intel, 2026a. Trend chart of TDP. www.intel.cn/ [accessed 25 March 2026].
Intel, 2026b. Purley. https://www.intel.cn/content/www/cn/zh/products/platforms/details/purley.html?ivk_sa=1024320u [accessed 21 July 2026].
Javidi, M., Entezari, M.H., 2024. The effect of ultrasonic waves on the structure, morphology, and thermal conductivity of graphene oxide as nanofluids for direct absorption solar collector application. Renew. Energ. 237, 121683.
Ji, Y., Zhou, X., Zhu, Y., Lu, X., Tuluhong, M., Zhang, H., Yang, Z., Zou, D., 2026. Immersion cooling technology: coolants and modification, cooling surface structure and system optimization. J. Energy Chem. 116, 91–122.
Jin, P., Yang, J., Zhang, X., Zhang, X., Xie, G., 2025. Heat transfer enhancement for high power chips in immersion cooling by ducted propellers. Appl. Therm. Eng. 273, 126469.
Jin, S., Yin, R., Li, Y., Tao, W., 2024. Numerical investigation on heat transfer performance of a single-phase immersion cooling system. In: 14th International Conference on Computational Heat and Mass Transfer.
Kelly, B., Moreno, G., Myers, S., Narumanchi, S., Joshi, Y., Graham, S., 2021. Dielectric fluids for the direct forced convection cooling of power electronics. In: 2021 20th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm).
Khoshvaght-Aliabadi, M., Ghodrati, P., Nasrolahzadeh, A., Kang, Y T, 2025. Optimization of heat sinks for data center server CPUs cooled via single-phase immersion cooling. Appl. Therm. Eng. 280, 127790.
Khoshvaght-Aliabadi, M., Hojjati, F., Hassani, S., 2026a. Influence of component arrangement on thermal management in immersion-cooled server boards. Energy 347, 140323.
Khoshvaght-Aliabadi, M., Nasrolahzadeh, A., Ghodrati, P., Kang, Y T, 2026b. Oblique-fin heat sink design for uniform CPU cooling in immersion-cooled servers. Appl. Therm. Eng. 288, 129628.
Kim, J., Choi, H., Lee, S., Lee, H., 2025. Computational study of single-phase immersion cooling for high-energy density server rack for data centers. Appl. Therm. Eng. 264, 125476.
Koibuchi, M., Fujiwara, I., Niwa, N., Totoki, T., Hirasawa, S., 2019. The case for water-immersion computer boards. In: Proceedings of the 48th International Conference on Parallel Processing.
Kong, R., Zhang, H., Tang, M., Zou, H., Tian, C., Ding, T., 2024. Enhancing data center cooling efficiency and ability: a comprehensive review of direct liquid cooling technologies. Energy 308, 132846.
Koster, D., Marongiu, A., Chahardahcherik, D., Braun, C F, Schulte, D., Figgemeier, E., 2023. Degradation analysis of 18650 cylindrical cell battery pack with immersion liquid cooling system. Part 1: aging assessment at pack level. J. Energy Storage 62, 106839.
Koutras, K N, Antonelou, A E, Naxakis, I A, Charalampakos, V P, Pyrgioti, E C, Yannopoulos, S N, 2022. In-situ high temperature study of the long-term stability and dielectric properties of nanofluids based on TiO2 and SiC dispersions in natural ester oil at various concentrations. J. Mol. Liq. 359, 119284.
Kurata, K., Hagihara, Y., Kuwata, M., Muto, T., Kobayashi, S., Pitwon, R., 2021. Silicon photonics transceiver evaluation for immersion cooled data center and HPC environments. In: 2021 IEEE 17th International Conference on Group IV Photonics.
Lamotte-Dawaghreh, J., Herring, J., Pundla, S A, Suthar, R., Nair, V., Bansode, P., Gupta, G., Agonafer, D., Madril, J., Ouradnik, T., Matthews, M., Winfield, I., 2024. Electrochemical additive manufacturing-based design of a heat sink for single-phase natural convection immersion cooling application. J. Electron. Packag. 146 (4).
Lawson A J, Offutt M C, Zhu L. Data centers and their energy consumption: frequently asked questions. 2026.
Li, J., Song, S., Shi, H., Xu, Y., Meng, E., Zhou, B., 2022. Thermal effects of water immersion on spray cooling in non-boiling zone. Int. Commun. Heat Mass Transf. 139, 106471.
Li, X., Guo, S., Sun, H., Liu, S., Wang, X., Wang, L., 2025. Comprehensive assessment of coolants used single-phase immersion cooling data center. Energy 340, 139240.
Li, X., Xu, Z., Liu, S., Zhang, X., Sun, H., 2023. Server performance optimization for single-phase immersion cooling data center. Appl. Therm. Eng. 224, 120080.
Li, X., Zou, C., Zhou, L., Qi, A., 2016. Experimental study on the thermo-physical properties of diathermic oil based SiC nanofluids for high temperature applications. Int. J. Heat. Mass Transf. 97, 631–637.
Liang, Y., Lin, X., Su, W., Xing, L., Li, W., Wang, B., 2023. Preliminary design and optimization of a solar-driven combined cooling and power system for a data center. Energy Convers. Manag.: X 20, 100409.
Liaw, D-J, Wang, K-L, Huang, Y-C, Lee, K-R, Lai, J-Y, Ha, C-S., 2012. Advanced polyimide materials: syntheses, physical properties and applications. Prog. Polym. Sci. 37 (7), 907–974.
Liu, C., Hao, N., Zhang, T., Wang, D., Li, Z., Bian, W., 2024a. Optimization of data-center immersion cooling using liquid air energy storage. J. Energ. Storage 90, 111806.
Liu, C., Huang, Y., Zhang, C., 2025a. Efficient jet-assisted single-phase immersion liquid cooling for high heat-flux servers. Appl. Therm. Eng. 259, 124935.
Liu, C-K, Chang, T-Y., 2022. Thermal performance of single-phase and two-phase immersion cooling in data center. In: 2022 17th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT).
Liu, J., Chen, H., Yang, M., Huang, S., Wang, K., 2024b. Comparative study of natural ester oil and mineral oil on the applicability of the immersion cooling for a battery module. Renew. Energ. 224, 120187.
Liu, S., Guo, S., Sun, H., Xu, Z., Li, X., Wang, X., 2025b. Evaluation of energy, economic, and pollution emission for single-phase immersion cooling data center with different economizers. Appl. Therm. Eng. 260, 125049.
Liu, S., Xu, Z., Wang, Z., Li, X., Sun, H., Zhang, X., Zhang, H., 2024c. Optimization and comprehensive evaluation of liquid cooling tank for single-phase immersion cooling data center. Appl. Therm. Eng. 245, 122864.
Lua, K B, Wang, C-C., 2026. Single-phase immersion cooling for data centers and HPC: recent advances and system-level challenges. Renew. Sustain. Energy Rev. 232, 116786.
Luo, Q., Wang, C., Wen, H., Liu, L., 2022. Research and optimization of thermophysical properties of sic oil-based nanofluids for data center immersion cooling. Int. Commun. Heat Mass Transf. 131, 105863.
Luo, Q., Wang, C., Wu, C., 2023. Study on heat transfer performance of immersion system based on SiC/white mineral oil composite nanofluids. Int. J. Therm. Sci. 187, 108203.
Lv, J., 2026. Intel's Granite Rapids and Sierra Forest will support 12-channel DDR5, with a TDP of up to 500W. https://www.expreview.com/87434.html [accessed 21 July 2026].
Mai, Z., Zhang, Z., Zhong, K., Wu, T., Wang, C., 2025. Synergistically enhanced stability and thermal properties of oil-based nanofluids through multi-scale surface grafting of HDTMS on al_2o_3 nanoparticles. Surf. Interfaces. 78, 108114.
Makaoui, A., Admi, Y., Moussaoui, M A, Mezrhab, A., 2025. Numerical study of influence of nanofluids on the optimization of heat transfer in immersion cooling systems. Processes 13 (3), 620.
Manzoor, U., Ramzan, M., Fida, M., Almehizia, A A, Khan, F A, Alatawi, A., 2026. Thermal and flow characteristics of Cu−TiO_2-diamond/water ternary hybrid nanofluids: applications in thermal energy storage systems. Diam. Relat. Mater. 167, 113865.
Matsuoka, M., Matsuda, K., Kubo, H., 2017. Liquid immersion cooling technology with natural convection in data center. In: 2017 IEEE 6th International Conference on Cloud Networking.
Min, K., Kang, T., Kang, T Y, Pyo, J-B., 2025. Evaluation of leakage currents of semiconductor packages due to high-voltage stress under an immersion cooling environment. Appl. Sci. 15 (9), 4668.
Ministry of Industry and Information Technology of the People's Republic of China. Technical requirements and test methods for cooling liquids of data center liquid cooling systems: YD/T 3982-2021. 2021-12-02.
Muneeshwaran, M., Lin, Y-C, Wang, C-C., 2023. Performance analysis of single-phase immersion cooling system of data center using FC-40 dielectric fluid. Int. Commun. Heat Mass Transf. 145, 106843.
National Development and Reform Commission of China, 2024. Special action plan for green and low-carbon development of data centers. https://www.gov.cn/zhengce/zhengceku/202407/content_6964165.htm [accessed 5 March 2026].
Nazari, M., Karami, M., Ashouri, M., 2014. Comparing the thermal performance of water, Ethylene Glycol, Alumina and CNT nanofluids in CPU cooling: experimental study. Exp. Therm. Fluid. Sci. 57, 371–377.
Ni, D., Yu, F., 2024. CFD simulation study of flow equalisation plate model in single-phase immersion liquid cooling for servers. Therm. Sci. Eng. Prog. 47, 102268.
Niazmand, A., Murthy, P., Saini, S., Shahi, P., Bansode, P., Agonafer, D., 2020. Numerical analysis of oil immersion cooling of a server using mineral oil and Al2O3 nanofluid. In: ASME 2020 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.
NVIDIA, 2026. Trend chart of TDP. www.nvidia.cn [accessed 25 March 2026].
Open Compute Project, 2022a. Base specification for immersion fluids. https://www.opencompute.org/documents/ocp-base-specification-for-immersion-fluids-20221201-pdf [accessed 4 March 2026].
Open Compute Project, 2022b. Material compatibility in immersion cooling. https://www.opencompute.org/documents/material-compatibility-in-immersion-cooling-document-version-1-0-nov-28-2022-1-pdf [accessed 2 March 2026].
Pambudi, N A, Sarifudin, A., Firdaus, R A, Ulfa, D K, Gandidi, I M, Romadhon, R., 2022. The immersion cooling technology: current and future development in energy saving. Alex. Eng. J. 61 (12), 9509–9527.
Pan, Y., Wang, H., Zong, H., Wang, P., Chen, H., Yu, H., 2024. Research progress on immersion liquid cooling technology for data center servers. Lubr. Oil 39 (06), 7–13.
Qian Y, Wang Q, Pan S, Zhao Y, Zhao Y. A single-phase immersion coolant, preparation method therefor, and application thereof: china, CN120865852A. 2025-10-31.
Rafati, M., Hamidi, A A, Shariati Niaser, M., 2012. Application of nanofluids in computer cooling systems (heat transfer performance of nanofluids). Appl. Therm. Eng. 45-46, 9–14.

Rahman, M A, Hasnain, S M M, Pandey, S., Tapalova, A., Akylbekov, N., Zairov, R., 2024. Review on nanofluids: preparation, properties, stability, and thermal performance augmentation in heat transfer applications. ACS. Omega 9 (30), 32328–32349.

Ramdas, S., Rajmane, P., Chauhan, T., Misrak, A., Agonafer, D., 2019. Impact of immersion cooling on thermo-mechanical properties of PCB's and reliability of electronic packages. In: ASME 2019 International Technical Conference and Exhibition On Packaging and Integration of Electronic and Photonic Microsystems.

Refai-Ahmed, G., Islam, M M, Kabbani, H., Cader, T., Alissa, H., Do, H., 2022. Holistic understanding of best engineering practice to extend cooling limit of next generations of GPU. In: 2022 IEEE 24th Electronics Packaging Technology Conference (EPTC).

Rehman, U U, Sahar, K U, Wang, C-M., 2026. Next-generation nanofluids: smart materials, hybrid composites, and the path to sustainable thermal management. J. Mol. Liq. 443, 129148.

Rokonuzzaman, A S M, Erdem, K., Şahin, B., Özdemir, M.R., 2025. Experimental study of a single-phase immersion cooling system with natural and forced convection. Int. J. Therm. Sci. 214, 109868.

Ruichek, A., Hnayno, M., Chehade, A., Fardoun, F., Moussa, T., Polidori, G., Maalouf, C., 2025. Immersion cooling fluid comparison using a single-phase immersion/liquid technique for data centers. Appl. Therm. Eng. 280, 128355.

Runhe Materials Technology Co. Ltd, 2025. RH-PCM-10 immersion silicone oil coolant. https://www.sekorm.com/doc/574388989.html [accessed 5 March 2026].

Shah, J M, Bhatt, C., Rachamreddy, P., Dandamudi, R., Saini, S., Dereje, A., 2019a. Computational form factor study of a 3rd generation open compute server for single-phase immersion cooling. In: ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.

Shah, J M, Eiland, R., Rajmane, P., Siddarth, A., Agonafer, D., Mulay, V., 2019b. Reliability considerations for oil immersion-cooled data centers. J. Electron. Packag. 141 (2).

Shah, J M, Eiland, R., Siddarth, A., Agonafer, D., 2015. Effects of mineral oil immersion cooling on IT equipment reliability and reliability enhancements to data center operations. In: 15th IEEE ITHERM Conference.

Shah, J M, Padmanaban, K., Singh, H., Duraisamy Asokan, S., Saini, S., Agonafer, D., 2021. Evaluating the reliability of passive server components for single-phase immersion cooling. J. Electron. Packag.

Shah, J M, Rizvi, S H I, Kota, I S, Nagilla, S R, Thakkar, D., Agonafer, D., 2016. Design considerations relating to non-thermal aspects of oil immersion cooling. In: ASME 2016 International Mechanical Engineering Congress and Exposition.

Shelton, J., Saini, N K, Hasan, S M, 2023. Experimental study of the rheological behavior of TiO2-Al2O3/mineral oil hybrid nanofluids. J. Mol. Liq. 380, 121688.

Shenyang Aojina Chemical Co. Ltd, 2026. Aojina data center immersion cooling fluid. http://www.original.com.cn/product/d704.html [accessed 5 March 2026].

Shrigondekar, H., Lin, Y-C, Wang, C-C., 2023. Investigations on performance of single-phase immersion cooling system. Int. J. Heat. Mass Transf. 206, 123961.

Singha, S., Laneryd, T., Gradinger, T., Lendenmann, H., 2017. Dielectric liquids for subsea high voltage equipment. In: 2017 IEEE 19th International Conference on Dielectric Liquids.

Sohel Murshed, S M, Nieto de Castro, C A, 2017. A critical review of traditional and emerging techniques and fluids for electronics cooling. Renew. Sustain. Energy Rev. 78, 821–833.

Solvay, 2024. A pioneering chemical company. https://www.solvay.com/en [accessed 2 March 2026].

Sristi, N A, Mim, S R, Billah, M M, Zaman, P B, 2025. Synthesis and performance evaluation of CuO/olive oil-based nanofluids: a systematic approach to developing stable suspension with optimal composition. J. Mol. Liq. 434, 128052.

Sun, X., Liu, Z., Ji, S., Yuan, K., 2025. Experimental study on thermal performance of a single-phase immersion cooling unit for high-density computing power data center. Int. J. Heat. Fluid. Flow. 112, 109735.

Sun, X., Liu, Z., 2026. Dynamic simulation and performance improvement on a single-phase immersion cooling system at cabinet level used for high computing-power data centers. Int. J. Heat. Mass Transf. 257, 128241.

Suresh, P., Vasista, K N, Kushwaha, S., Balaji, C., 2024. Thermal management of multi-chip electronic module with wavy fin heat sink-assisted immersion cooling. J. Phys.: Conf. Ser. 2766 (1), 012182.

Suthar, R., Bansode, P., Venkatachalam, V V T, Ede, N T, Lakshminarayana, A., Bhandari, R., Sivakumar, A., Agonafer, D., 2024. Investigating the effect of single-phase liquid immersion cooling on TG-400 substrate core's thermomechanical characteristics. In: ASME 2024 International Mechanical Engineering Congress and Exposition.

Taddeo, P., Romaní, J., Summers, J., Gustafsson, J., Martorell, I., Salom, J., 2023. Experimental and numerical analysis of the thermal behaviour of a single-phase immersion-cooled data centre. Appl. Therm. Eng. 234, 121260.

Thomas, S., Sobhan, C B, Taha-Tijerina, J., Narayanan, T N, Ajayan, P M, 2012. Investigations on transient natural convection in boron nitride-mineral oil nanofluid systems. In: ASME 2012 International Mechanical Engineering Congress & Exposition.

Umar, S., Sulaiman, F., Abdullah, N., Mohamad, S.N., 2020. Preparation, stability and thermal characteristic of al_2o_3/bio-oil based nanofluids for heat transfer applications. J. Nanosci. Nanotechnol. 20 (12), 7569–7576.

Vagiakis, A., Korres, D N, Tzivanidis, C., 2025. Simulation of natural convection heat transfer in dielectric liquids for single-phase immersion cooling rack server. Appl. Therm. Eng. 274, 126595.

Van Vaerenbergh, B., Kerbrat, M., Struelens, P., 2025. A novel functional accelerated thermal aging methodology for single-phase dielectric immersion coolants. ACS. Omega 10 (43), 50882–50895.

Wang, C., Yu, H., Xian, L., Liu, G., Chen, L., Tao, W-Q., 2024a. Study on the operation strategy of the water cooling system in the single-phase immersion cooling data center considering cooling tower performance. Energy Build. 324, 114890.

Wang, H., Yuan, X., Zhang, K., Lang, X., Chen, H., Yu, H., Li, S., 2024b. Performance evaluation and optimization of data center servers using single-phase immersion cooling. Int. J. Heat. Mass Transf. 221, 125057.

Wang, N., Guo, Y., Huang, C., Shao, S., 2024c. Advances in direct liquid cooling technology and waste heat recovery for data center: a state-of-the-art review. J. Clean. Prod. 477, 143872.

Wang, Y., Gui, C., Cheng, P., Shen, C., Wang, R., Lv, Z., Lin, B., Tian, W., Li, Z., Zhang, X., Zhang, K., Wu, L., Ahuja, N., 2023a. Investigation on Immersion cooling solution for hyper-scale data center application. In: 2023 22nd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm).

Wang, Y., Gui, C., Shen, C., Cheng, P., Wang, C., Wang, R., Wang, S., Jing, T., Lin, B., Tian, W., Xu, J., Li, Z., Zhang, Z., Zhang, X., Zhang, Z., Wu, L., Wang, K., Ahuja, S., Ahuja, N., 2024d. Analysis on long term reliability of single-phase immersion solution based on data center deployment. In: 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm).

Wang, Y., Tang, Z., Niu, F., Huang, Q., Song, H., Xu, Y., 2025. 3D simulation calculation and analysis of winding temperature distribution in a 220 kV natural ester transformer. Insul. Mater. 58 (8), 50–56.

Wang, Z., Zhao, R., Wang, S., Huang, D., 2023b. Heat transfer characteristics and influencing factors of immersion coupled direct cooling for battery thermal management. J. Energy Storage 62, 106821.

Wei, B., Zou, C., Yuan, X., Li, X., 2017. Thermo-physical property evaluation of diathermic oil based hybrid nanofluids for heat transfer applications. Int. J. Heat. Mass Transf. 107, 281–287.

Yang, C., Zhu, X., Zhang, G., Pan, G., Wang, X., 2026. A review of the immersion liquid cooling technology for high-performance data centers. Int. J. Heat. Fluid. Flow. 119, 110282.

Yu, L., Yang, J., Shia, D., Zhang, M., 2022. A study on compatibility of thermal interface materials with coolants for data center immersion cooling. In: 2022 21st IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm).

Yuan, L., Wang, Y., Kosonen, R., Yang, Z., Zhang, Y., Wang, X., 2024. Comparative study on heat dissipation performance of pure immersion and immersion jet liquid cooling system for single server. Buildings 14 (9), 2635.

3M, 2019. 3M™ Fluorinert™ Electronic Liquid FC-70. https://multimedia.3mcom/mws/media/64891O/fluorinert-electronic-liquid-fc-70.pdf [accessed 4 March 2026].

Zhang, B., Li, H., Xu, T., Wang, L., Chen, L., Li, Z., 2026. Integrated CFD modeling and experimental validation of single-phase immersion cooling for data center thermal management. Energy 347, 140315.

Zhang C, Guo Q, Jia X, Quan H. The fluorine-containing heat transfer fluid with cyclic skeleton, preparation method and application thereof: china, CN112094627B. 2021-03-16.

Zhang, C., Guo, Q., Jia, X., Quan, H., 2022a. Research progress of immersion coolant for data centers. Fine Chem. 39 (11), 2184–2195.

Zhang C, Guo Q, Jia X, Quan H. Cyclic skeleton fluorine-containing heat transfer fluid as well as preparation method and application thereof: china, CN112980396B. 2022b-04-12.

Zhang, C., Wang, H., Huang, Y., Zhang, L., Chen, Y., 2025a. Immersion liquid cooling for electronics: materials, systems, applications and prospects. Renew. Sustain. Energy Rev. 208, 114989.

Zhang, X., Zhu, H., Feng, S., Yang, Y., Chen, C., Zhao, X., Yang, L., 2025b. Research on generation mechanism of carbon particles in natural ester and mineral oil and their impact on electrical properties. High Volt. Appar. 61 (10), 0011–0020.

Zhang, Y-D, Lin, Y-C, Wang, C-C., 2024. Investigation of the single-phase immersion cold plate amid PAO-4 and Noah@3000A – An experimental approach and its numerical verification. Int. Commun. Heat Mass Transf. 155, 107509.

Zhejiang Noah Fluorochemical Co. Ltd, 2025a. Noah® 3000EP single-phase electronic coolant. https://www.zjnoah.cn/index.php?m=content&c=index&a=show&catid=65&id=3 [accessed 18 January 2026].

Zhejiang Noah Fluorochemical Co. Ltd, 2025b. Noah® Coolant. https://www.zjnoah.cn/index.php?m=content&c=index&a=lists&catid=65.

ZheJiang Yonghe Refrigerant Co. Ltd, 2025. ICELOONG® Trimeric 3123A single-phase electronic coolant. https://www.qhyh.com/index.php/Goods/goods_x/article_id/2241.html [accessed 5 March 2026].

Zheng, J., Chen, X., Cheng, B., He, Y., Tian, W., Zhong, Y., Yu, J., Zhang, Z., He, Y., Li, Y., Zhang, Z., Wu, L., Ahuja, N., 2024. Experimental investigation on single-phase immersion cooling solution for data center application. In: 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm).

Zheng, S., Su, C., Yang, X., Zhang, Y., Duan, K., Zhang, Y., Huang, Z., Zhang, Y., Liu, F., Wei, J., 2025. A comprehensive review of single-phase immersion cooling in data centres. Appl. Therm. Eng. 272, 126385.

Zhou, K., Yu, X., Xie, B., Xie, H., Fu, W., Liao, Q., 2024. Immersion cooling technology development status of data center. Sci. Technol. Energy Transit. 79, 41.

Zhu, X., Lu, T., Guan, X., Tian, Z., Cheng, H., Fang, Z., 2026. Atmospheric pressure plasma surface treatment to enhance interfacial insulation and water resistance of PP/SIR composite insulation. Appl. Surf. Sci. 717, 164790.

Zou, S., Zhang, J., Wang, Y., Xiao, T., Huang, J., 2025. Multi-objective optimization of server component temperatures and coolant filling ratio in single-phase immersion tank based on Taguchi grey correlation analysis. Case Stud. Therm. Eng. 72, 106363.